

CLINOMA PLATFORM

presents

SECOND PAPER EDITION

- ☐ CHEST DISEASES
- ☐ NEONATOLOGY
- ☐ RENAL DISEASES
- ☐ EMERGENCY MEDICINE
- ☐ FAMILY MEDICINE



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Tip: Use the interactive landscape view on your tablet or laptop for optimal clinical visualization.

CHAPTER 01

I. Chest Diseases



**VIRAL CROUP • EPIGLOTTITIS • DD CROUP & PNEUMONIA • BRONCHIOLITIS • BRONCHIAL ASTHMA • WHEEZING & FOREIGN BODY • CHILDHOOD PNEUMONIA:
CLASSIFICATIONS • CHILDHOOD PNEUMONIA: MISLEADING SIGNS • CHILDHOOD PNEUMONIA: DIAGNOSTICS & ICU • CHILDHOOD PNEUMONIA: THERAPEUTICS • PNEUMONIA
SUMMARY**

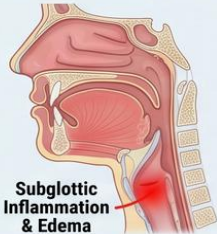
PEDIATRIC VIRAL CROUP: COMPREHENSIVE CLINICAL PICTURE, DIAGNOSTICS & EMERGENCY PROTOCOLS

A high-yield medical guide focusing on **laryngo-tracheo-bronchitis** in children.

Definition, Epidemiology & Microbial Etiology

Croup, or **laryngo-tracheo-bronchitis**, is an acute viral infection of the glottic and subglottic regions. It stands as the most common of the inflammatory upper airway obstruction syndromes in children. It primarily affects younger children during the fall and early winter seasons.

Primary Cause: Driven by respiratory viruses, most commonly Parainfluenza virus types 1 and 3.
Other Causes: Includes Respiratory Syncytial Virus RSV, influenza virus, rubeola virus, adenovirus, and Mycoplasma pneumoniae.



Fall & Early Winter Seasons

Parainfluenza Virus Types 1 & 3 (Most Common)

RSV, Influenza, Adenovirus, M. pneumoniae (Other Causes)

Clinical Progression & Impending Respiratory Failure Flags

Begins with a typical **prodrome** of an upper respiratory tract infection, followed by a characteristic **barking cough**, **inspiratory stridor**, **hoarseness of voice**, and variable respiratory distress. Fever is usually low grade. Symptoms are characteristically worse at night, recur for several days, and become more severe when the child cries.

Critical Red Flags: Stridor at rest (without crying) indicates that croup is more severe.

Physical examination may reveal **decreased air entry**, **rhonchi**, and **scattered crackles**.

Signs of impending respiratory failure include central cyanosis, pallor, and a decreased level of consciousness.

Crucially, stridor, barking cough, and chest retractions may not be severe if the child's respiratory effort is failing due to total exhaustion.

Croup is diagnosed clinically, particularly by observing the barking cough and stridor. **Neck X-ray** is usually not needed for diagnosis.
Postero-anterior view: shows the pathognomonic steeple sign, reflecting uniform subglottic narrowing of the airway. **Lateral view:** done primarily to evaluate supraglottic structures and exclude acute epiglottitis.



Barking Cough & Stridor



Symptoms Worse at Night, Recur for Days, Exacerbated by Crying

CRITICAL RED FLAG: Stridor at Rest

Terminal Failure Parameters:

Central Cyanosis, Pallor, Decreased Level of Consciousness

Warning: Failing Effort May Mask Severity Due to Exhaustion.



Narrowed Subglottic Airway

STEEPLE SIGN (Pathognomonic)

Clinical Diagnosis Confirmed by Barking Cough & Stridor.

Neck X-ray Usually Not Needed.

Diagnostic Rationale & Structural Neck Radiography

Croup is diagnosed clinically, particularly by observing the **barking cough** and **stridor**.

Neck X-ray is usually not needed for diagnosis.

Postero-anterior view: shows the pathognomonic steeple sign, reflecting uniform subglottic narrowing of the airway.

Lateral view: done primarily to evaluate supraglottic structures and exclude acute epiglottitis.

Narrowed Subglottic Airway



STEEPLE SIGN (Pathognomonic)

Clinical Diagnosis Confirmed by Barking Cough & Stridor.

Neck X-ray Usually Not Needed.

Emergency Management Protocol & Strict Restrictions

Treatment of respiratory distress takes priority over any testing.

Emergency lines:

1. **Supplemental Oxygen** to alleviate hypoxemia.
2. **Nothing per mouth (NPO).**
3. **Corticosteroids:** a single dose of Dexamethasone given as 0.6 mg/kg IM or oral to reduce mucosal inflammation.
4. **Epinephrine:** decreases airway edema.

Nebulized L-epinephrine (5 mL of 1:1,000 solution) or racemic epinephrine can be used. Its duration of action is 2 hours, so observe the patient for recurrence/rebound symptoms. Bag and mask ventilation or **endotracheal intubation** may be needed in severe cases; most patients recover readily with steroids and epinephrine.

Strict Prohibitions: Avoid agitating the child, e.g., by blood tests or IV lines. Absolutely avoid using a tongue depressor as it may cause sudden cardiac arrest.



L-Epinephrine 1:1,000 (5 mL)



Observe for Recurrence/Rebound Symptoms: Duration of Action 2 Hours

Severe Cases
Bag mask
Endotracheal intubation



STRICTLY PROHIBITED: Avoid Tongue Depressor – May Cause Sudden Cardiac Arrest! Avoid Agitation.

High-Yield Variant: Spasmodic Croup

Spasmodic Croup is a brief, less severe variant occurring mostly in children 1-3 years old. Usually occurs suddenly in the evening or night, where the child awakens with a barking cough, noisy inspiration, and respiratory distress. Resolves within several hours. Pathogenesis is unknown (viral or allergic). It is clinically similar to acute laryngotracheobronchitis, except for three distinct features:

1. No viral prodrome.
2. No fever.
3. No family members with respiratory infections.



Occurs Suddenly at Night, Resolves Within Hours

Aged 1-3 Years

NO Viral Prodrome

NO Fever

NO Infected Family Contacts

PEDIATRIC ACUTE EPIGLOTTITIS: CLINICAL OVERVIEW & MANAGEMENT

A Life-Threatening Upper Airway Obstruction

1. CORE DEFINITION & SPECIFIC MICROBIAL ETIOLOGY

Acute epiglottitis is a **rapidly progressive, life-threatening infection** of the epiglottis causing **acute upper airway obstruction**.

Its incidence has dramatically decreased after routine vaccination for **Hemophilus influenzae bacteria**.

The underlying etiological agent is highly specific, caused by **Hemophilus influenzae type b** in almost all cases.



2. AGE DISTRIBUTION & SYSTEMIC SYMPTOMS

The disease manifests with a rapid onset of **severe systemic signs**.

Age Group: Classically affects children from 2 to 7 years of age, with a sharp clinical peak at 3.5 years.

Presents with an abrupt onset of **high-grade fever** and a **severe, acute sore throat**.



4. DIRECT DIAGNOSIS CONFIRMATION & AIRWAY SECURITY



Diagnosis is established by **directly visualizing a swollen, cherry-red epiglottis** via a **laryngoscope**.

Crucially, this examination should be performed only in an operating room or intensive care unit by a skilled doctor.

Insertion of an **artificial airway**, usually an **endotracheal tube** or less often tracheostomy, must be performed immediately to all patients having epiglottitis in an operating room or intensive care unit by a skilled doctor.

This proactive procedure has successfully reduced the disease mortality to zero.

The endotracheal tube is **safely removed after a few days** when direct laryngoscopy shows that the supraglottic edema has completely subsided.



3. THE FOUR PATHOGNOMONIC HALLMARKS & SURVIVAL WARNING

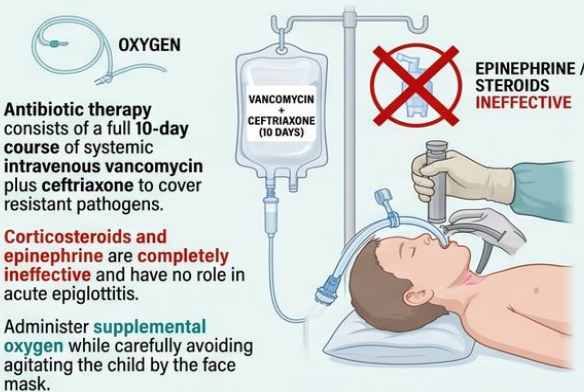
As supraglottic inflammation intensifies, **complete obstruction of the airway and death can occur rapidly**.

The clinical diagnosis is driven by **four classic hallmarks**:

- 1. Drooling of Saliva** reflecting severe dysphagia and an absolute inability to swallow.
- 2. Hyperextension of the Neck** where the child sits upright leaning forward (**tripod position**) to mechanically maintain airway patency.
- 3. Respiratory Distress** presenting with **rapid, shallow breathing**.
- 4. Absence of Barking Cough** which is characteristically completely rare, while **stridor** stands as a late and ominous sign of near-total occlusion.



5. TARGETED MIDICAL THERAPEUTICS & INEFFECT



6. CRITICAL GENERAL RESTRICTIONS & ABSOLUTE EXAMINATION BANS

During management, the clinician must strictly adhere to the following safety protocols:

- 1. Absolutely avoid direct inspection of the oral cavity by a tongue depressor** as it may incite a severe vasovagal reflex or **total laryngospasm**, causing immediate, complete airway occlusion and sudden cardiac arrest.
- 2. Avoid placing the child in a supine position** as gravity causes the swollen epiglottis to drop backward and seal the airway.
- 3. Avoid agitating the child** by phlebotomy or intravenous line placement before airway control as crying increases turbulent airflow.
- 4. Avoid the administration of sedation, opiate derivatives, expectorants, and antihistamines.**



DIFFERENTIAL DIAGNOSIS: CROUP SYNDROMES vs. CHILDHOOD PNEUMONIA

CROUP SYNDROMES (DIFFERENTIAL)



Diphtheritic croup

Pharyngeal examination reveals typical **gray-white membrane**.



Measles croup

Coincides with systemic manifestations of **measles**.



Aspiration of a foreign body

Choking and coughing occur suddenly, **without prodromal signs** of infection.



Retropharyngeal or peritonsillar abscess



Extrinsic compression of the airway

Hematoma from trauma or vascular ring.



Intraluminal obstruction

Laryngeal **papilloma** or subglottic hemangioma.



Angioedema of the subglottic areas

Part of **anaphylaxis**.



Hypocalcemic tetany



Infectious mononucleosis



Trauma



Malformations of the larynx



Vocal cord dysfunction

Paradoxical closure of the **vocal cords** during inspiration diagnosed by laryngoscopy.



Tumors



Accidental ingestion of very hot liquid

Leads to **dysphagia**, **drooling**, and **stridor**.



Bronchial asthma

A croupy cough may be an early sign of **asthma**.

CHILDHOOD PNEUMONIA (DIFFERENTIAL)



Diabetic Ketoacidosis (DKA)

Causes **tachypnea** but breathing is **deep** (Kussmaul's), whereas it is shallow in pneumonia.



Upper airway disorders



Lower airway disorders



Foreign body



Non-pneumonic pleural effusion



Mediastinal mass



Congenital lung anomalies

Sequestration, **CCAM** causing recurrent pneumonia in the same site.



Congenital lung anomalies

Sequestration, **CCAM** causing recurrent pneumonia in the same site.



Heart failure

Murmur, gallop, tender liver, **cardiomegaly** on chest X-ray, low EF on echo.



Acute Kidney Injury (AKI)

Fluid overload, metabolic acidosis.



Central Nervous System (CNS) insults

Intracranial hemorrhage causing central hyperventilation.



Abdominal pain mimicking surgical conditions



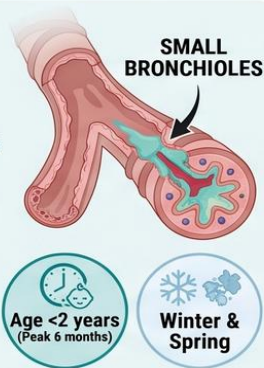
Chemical pneumonitis

ACUTE BRONCHIOLITIS: PATHOLOGY, CLINICAL PICTURE & SUPPORTIVE CARE

1. CORE DEFINITION & EPIDEMIOLOGICAL PARAMETERS

Acute Bronchiolitis is defined as the **obstruction of small airways by mucus and inflammatory cells**. It occurs during the first 2 years of life with a **peak incidence at 6 months** of age. Cases demonstrate a distinct **seasonal peak** during **winter and spring**.

The primary causative agent is **Respiratory Syncytial Virus RSV**, responsible for **50% of cases**. The remaining 50% is driven by other organisms, including Parainfluenza, Mycoplasma, Adenovirus, Herpes, and Rhinovirus.

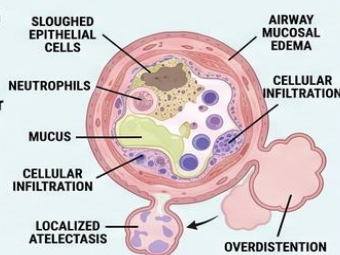


2. PATHOGENESIS & AIRWAY OBSTRUCTION MECHANISM

Respiratory syncytial virus infection results in a **loss of epithelial cilia** and **sloughing of epithelial cells** in the airways, leading to a collection of **desquamated airway epithelial cells, polymorphonuclear cells, and lymphocytes** within the airway. **Cellular infiltration and airway mucosal edema** occur with very little or no alveolar infiltration.

Sloughed epithelial cells, neutrophils, and lymphocytes are the major contributors to **airway obstruction**.

Complete plugging of some airways and partial plugging of others leads to **localized atelectasis** of some units of lung parenchyma and **overdistention** of other units, resulting in a **ventilation-perfusion mismatch** causing **hypoxemia**, which is generally relieved by **oxygen**.



VENTILATION-PERFUSION MISMATCH

HYPOXEMIA (RELIEVED BY OXYGEN) BY OXYGEN)

3. CLINICAL PRESENTATION PROFILE & DIAGNOSTIC FINDINGS

Features a history of **exposure to older children and adults with minor respiratory disease**, followed a week later by an **upper respiratory tract infection** with nasal discharge and sneezing.

Clinical findings include:

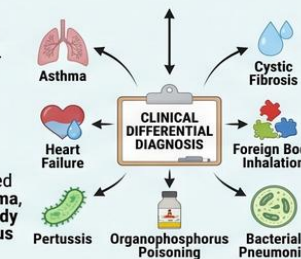
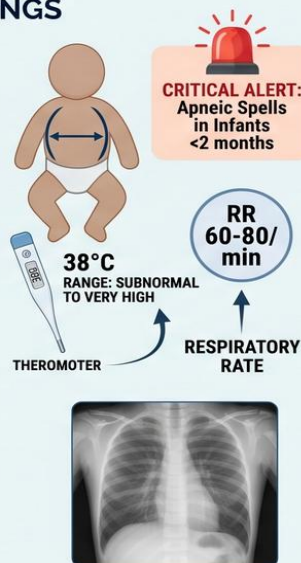
1. **Fever 38°C**, with a temperature ranging from **subnormal to very high**.
2. **Dry repetitive cough** or sharp dry cough.
3. **Respiratory distress and tachypnea** with a respiratory rate RR of 60-80/min.
4. **Central cyanosis** or **pallor**, and signs of **dehydration**.
5. **Increased anteroposterior diameter** of the chest, making it **hyper-resonant** on percussion.
6. **Widespread fine consonating crepitations** and **expiratory rhonchi** on auscultation, with a characteristically **prolonged expiration**.

Family history is characteristically **negative for atopy**.

Young infants below 2 months of age (or <4 months) are at **high risk** for presenting with **sudden apneic spells**.

Laboratory evaluation shows a **normal** or only slightly raised leukocytic count. **Chest radiograph demonstrates a hyper-translucent lung** secondary to **over-inflation**; in about 1/3 of cases, **localized consolidation** due to atelectasis or inflammation of alveoli is observed.

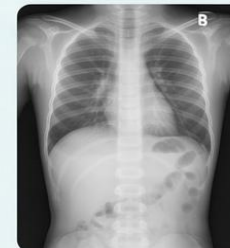
True bronchiolitis must be differentiated from clinically similar conditions: **asthma, cystic fibrosis, heart failure, foreign body inhalation, pertussis, organophosphorus poisoning, and bacterial pneumonia**.



4. LABORATORY, RADIOGRAPHY & DIFFERENTIAL DIAGNOSIS

Laboratory evaluation shows a **normal** or only **slightly raised leukocytic count**. **Chest radiograph demonstrates a hyper-translucent lung** secondary to **over-inflation**; in about 1/3 of cases, **localized consolidation** due to **atelectasis or inflammation of alveoli** is observed.

True bronchiolitis must be differentiated from clinically similar conditions: **asthma, cystic fibrosis, heart failure, foreign body inhalation, pertussis, organophosphorus poisoning, and bacterial pneumonia**.



5. THERAPEUTIC PROTOCOLS, COMPLICATIONS & STRICT BANS



ABSOLUTELY AVOID: NO STEROIDS, NO SEDATIVES, NO TRACHEOSTOMY

Clinical complications feature **secondary bacterial infections, otitis media, pneumothorax, apneic spells, anemia, dehydration, and cardiorespiratory collapse or heart failure**.

Treatment requires:

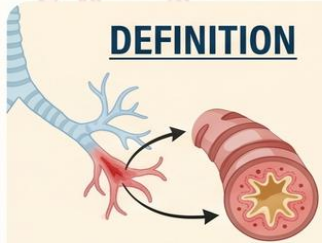
1. **Hospitalization** if distressed.
2. **Humidified oxygen** to relieve hypoxemia.
3. **Parenteral fluids** via intravenous infusion to maintain electrolyte balance and perform proper pH adjustment.
4. Consider a specific **antiviral agent Ribavirin**.
5. Administer a **bronchodilators** single-dose trial.

Antibiotics have no therapeutic role unless a **secondary bacterial infection** occurs. Provide **mechanical ventilation** in case of **respiratory failure**.

Strict Contraindications: Absolutely avoid sedatives, steroids, or tracheostomy.

BRONCHIAL ASTHMA: CLINICAL PICTURE, TRIGGERS, & INVESTIGATION

DEFINITION



Asthma is a diffuse obstructive lung disease with:

Hyperactivity of airways to variety of stimuli

A high degree of reversibility of the obstruction either spontaneously or with treatment

Chronic inflammation of the airways

ASTHMA TRIGGERS & COMORBIDITIES

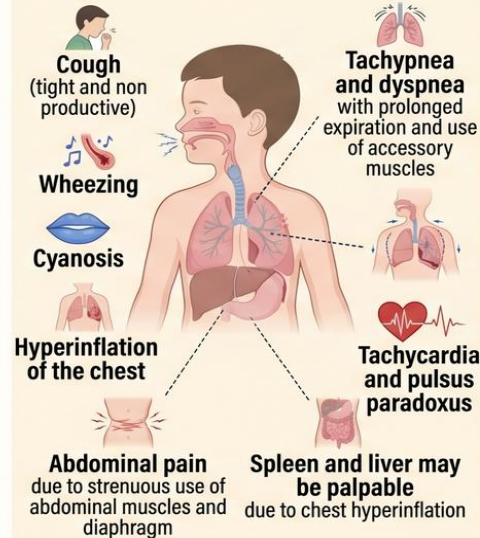
- **Viral respiratory infections**
- **Indoor Allergens** e.g. Animal dander, dust mites, cockroaches
- **Seasonal allergens** e.g. Pollens
- **Air pollutants** e.g. tobacco smoke, dust
- **Strong odors** e.g. Perfumes, hairsprays, Cleaning agents
- **Occupational exposure** e.g. Farm exposures, paint fumes
- **Drugs** e.g. aspirin and other NSAIDs, Beta-Blockers
- **Cold dry air**
- **Exercise**
- **Crying, laughter, hyperventilation**

Asthma comorbid conditions (conditions commonly associated with asthma and make asthma difficult to treat)

- 1- Rhinitis
- 2- Sinusitis
- 3- Gastroesophageal reflux

CLINICAL PICTURE

Acute episodes are caused by exposure to irritants e.g. cold air, noxious fumes () while **insidious onset** occurs following viral infections



Between exacerbations the child may be completely free of symptoms and signs.

INVESTIGATION & DIAGNOSIS

1- History: past history of recurrent wheezy chest and family history of asthma or other allergic disorders

2- Clinical picture

3- Laboratory investigations:

Blood examination:

- CBC: usually shows **eosinophilia** more than 250-400 cell/mm³
- Serum IgE levels are usually elevated

Sputum: white and tenacious. Usually there is **eosinophilia** and granules from disrupted cells

Pulmonary function tests:

- Done by **Spirometry** for children > 5 years
- **Low FEV1**
- **Low FEV1/FVC ratio**
- **Increase in FEV1 >9-12% after inhalation of short-acting β-agonist** indicates reversible air flow limitation
- **Worsening in FEV1 ≥15% after exercise challenge**
- **Day-to-day and/or morning-to-evening variation of PEF or FEV1 ≥20%**

Chest X ray: is required only to exclude other possible diagnosis or complications

Blood gas analysis:

- Important during exacerbations requiring hospitalization
- **PCO2:** usually low in early stages but high as obstruction worsens
- **PO2:** is low
- **pH** remains normal until late acidosis develops

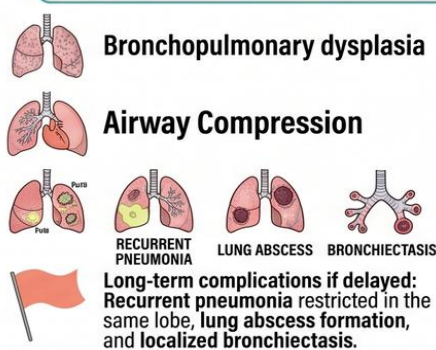
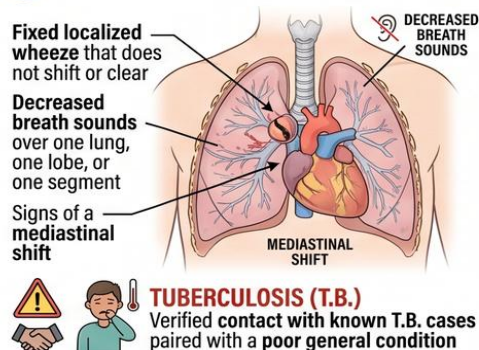
ACUTE & RECURRENT WHEEZING: HISTORY, DIFFERENTIAL DIAGNOSIS & FOREIGN BODY ASPIRATION

1 CLINICAL CAUSES MATRIX OF ACUTE vs. RECURRENT WHEEZING

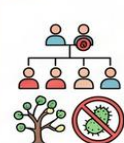
A) ACUTE WHEEZING	B) RECURRENT WHEEZING
 Acute Bronchiolitis	 Bronchial Asthma
 Severe Bacterial Bronchopneumonia	 Recurrent Aspiration (Infants with GERD where gastric contents enter the esophagus then the respiratory tract, causing inflammation)
 Foreign Body Aspiration	 Chronic Chest Infection
 First Episode of Asthma	 Acute Congestive Heart Failure or Recurrent/Chronic Heart Failure
 Organophosphorus Poisoning	

→ Categorized by Clinical Onset

3 PHYSICAL EXAMINATION SIGNS



2 HIGH-YIELD DIAGNOSTIC CLUES FROM HISTORY TAKING



ASTHMA

Recurrent wheezy chest in the **absolute absence of a viral prodrome**, displaying a clear relation to **allergens or physical exercise**, accon a **positive family history** of relatives with asthma.



ACUTE BRONCHIOLITIS

Limited to an **age below 2 years**, with a **mandatory history of a preceding upper respiratory catarrh**.



FOREIGN BODY ASPIRATION

Common in children **<4 years old** and in the **mentally retarded**, featuring a **sudden onset of choking** or a **frank, witnessed history of foreign body aspiration**.



TUBERCULOSIS (T.B.)

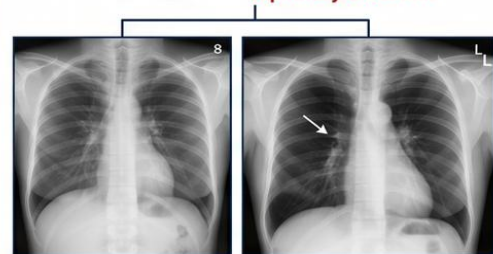
Verified **contact with known T.B. cases** paired with a **poor general condition**.

4 RADIOGRAPHIC TRACKING & THERAPEUTIC PROTOCOL

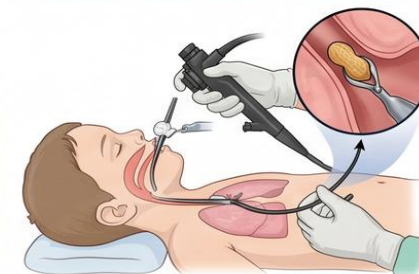
X-ray findings are **positive only in about 50%** of foreign body aspiration cases, as many objects are **radiolucent**.

Treatment requires addressing the underlying cause and **foreign foreign inoves**.

The radiograph may demonstrate **obstructive collapse (atelectasis)** or **obstructive emphysema**, which is best visualized in an **expiratory chest film**.



EXPIRATORY CHEST FILM (VISUALIZATION OF OBSTRUCTIVE EMPHYSEMA) Unilateral hyperinflation (obstructive emphysema) due to unseen foreign object



THERAPEUTIC PROTOCOL (BRONCHOSCOPIC EXTRACTION)

Treatment requires addressing the underlying cause and **immediate bronchoscopic extraction** of the foreign body.

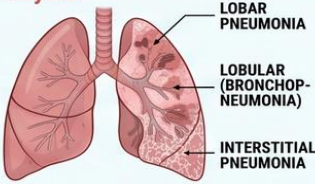
CHILDHOOD PNEUMONIA: CLASSIFICATION & DIAGNOSTIC CRITERIA (PART 1)

SECTION 1: CORE DEFINITIONS & CLASSIFICATIONS

Pneumonia is the **inflammatory inflammation of lung parenchyma**.
Anatomically: Classified into **lobar**, **lobular (bronchopneumonia)**, or **interstitial**.

Clinical Settings:

- Community-Acquired CAP:** Acquired outside hospital; **leading cause of death globally in children <5 years**.
- Hospital-Acquired HAP:** Nosocomial infection presenting **>=48 hours** after admission.
- Ventilator-Associated VAP:** Presents **>48 hours** after endotracheal intubation.



CAP ACQUIRED OUTSIDE HOSPITAL, LEADING GLOBAL CAUSE OF DEATH <5Y



HAP >=48 HRS AFTER ADMISSION (NOSOCOMIAL)



VAP >48 HRS AFTER ENDOTRACHEAL INTUBATION

SECTION 2: COMPREHENSIVE ETIOLOGY SPECTRUM

- Gram-Positive Bacteria:** *S. pneumoniae*, *S. aureus*, Group A/B *Streptococci*.
- Gram-Negative Bacteria:** *E. coli*, *H. influenzae*, *K. pneumoniae*.
- Atypical Bacteria:** *Mycoplasma* and *Chlamydia pneumoniae*.
- Viral:** RSV, adenovirus, influenza, and SARS-CoV-2.
- Fungal/Parasitic:** *Aspergillus*, *Pneumocystis jirovecii*, *Ascaris*.
- Non-Infectious Triggers:** Chemical pneumonitis (kerosene, hydrocarbons, meconium), gastric aspiration (GERD), hypersensitivity (hay/dust), and hypostatic pneumonia. Löffler syndrome features a transient pulmonary eosinophilic infiltrate from parasites or drugs.

GRAM-POSITIVE



GRAM-NEGATIVE



NON-INFECTIOUS TRIGGERS



ATYPICAL



VIRAL



CHEMICAL PNEUMONITIS (KEROSENE)



GASTRIC ASPIRATION (GERD)



VIRAL



FUNGAL/ PARASITIC



MECONIUM



HYPERSENSI (POLLEN) TRIGGERS



SECTION 3: CLINICAL PRESENTATION & WHO FAST BREATHING GRID

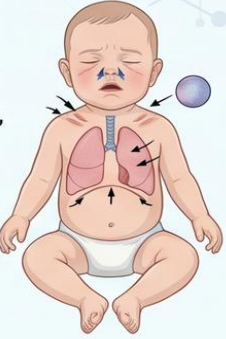
Presents with **high-grade fever**, **cough**, and **chest pain**.

Tachypnea is the most consistent and sensitive sign.

WHO Fast Breathing Grid (counted for **1 full minute at rest**):

AGE BRACKET	FAST BREATHING CUT-OFF
Birth to 2 months	>60 breaths/min
2 months to 1 year	>50 breaths/min
1 year to 5 years	>50 breaths/min
1 year to 5 years	>40 breaths/min

Distress Signs: Grunting, nasal flaring, chest retractions (intercostal/subcostal), and central cyanosis.

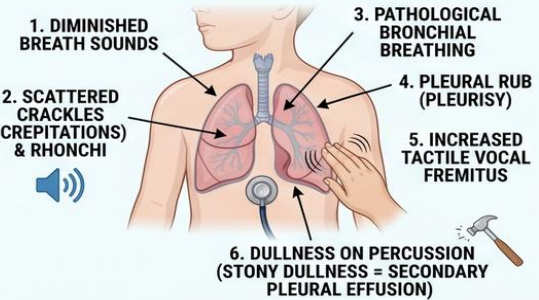


In young infants, physical examination can be misleading with **severe tachypnea** but **mild chest findings**.

SECTION 4: LOCAL CHEST PHYSICAL EXAMINATION SIGNS

Local signs over the affected lung field include:

- Diminished breath sounds.**
- Scattered crackles** (crepitations) and **rhonchi**.
- Pathological bronchial breathing.**
- Pleural rub** (indicates pleurisy).
- Increased tactile vocal fremitus.**
- Dullness on percussion** (stony dullness indicates secondary pleural effusion).



CHILDHOOD PNEUMONIA: MISLEADING NON-RESPIRATORY MANIFESTATIONS & ETIOLOGICAL CLUES (PART 2)

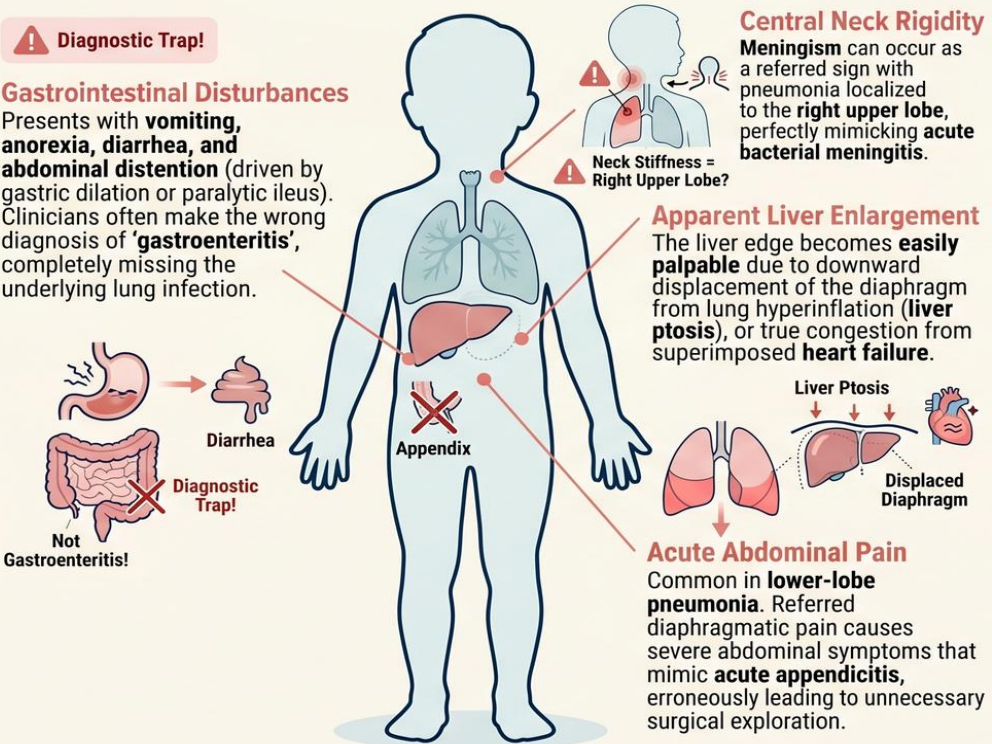
Explanatory Graphic - Academic Medical Structure

SECTION 1: MISLEADING NON-RESPIRATORY MANIFESTATIONS (The Clinical Traps)

Pneumonia in children can present with severe systemic, non-respiratory signs that mimic other non-pulmonary diseases, leading to dangerous diagnostic errors:

⚠ Diagnostic Trap!

Gastrointestinal Disturbances
Presents with **vomiting, anorexia, diarrhea, and abdominal distention** (driven by gastric dilation or paralytic ileus). Clinicians often make the wrong diagnosis of **'gastroenteritis'**, completely missing the underlying lung infection.



Central Neck Rigidity
Meningism can occur as a referred sign with pneumonia localized to the **right upper lobe**, perfectly mimicking **acute bacterial meningitis**.

⚠ Neck Stiffness = Right Upper Lobe?

Apparent Liver Enlargement
The liver edge becomes **easily palpable** due to downward displacement of the diaphragm from lung hyperinflation (**liver ptosis**), or true congestion from superimposed **heart failure**.

Liver Ptosis
Displaced Diaphragm

Acute Abdominal Pain
Common in **lower-lobe pneumonia**. Referred diaphragmatic pain causes severe abdominal symptoms that mimic **acute appendicitis**, erroneously leading to unnecessary surgical exploration.

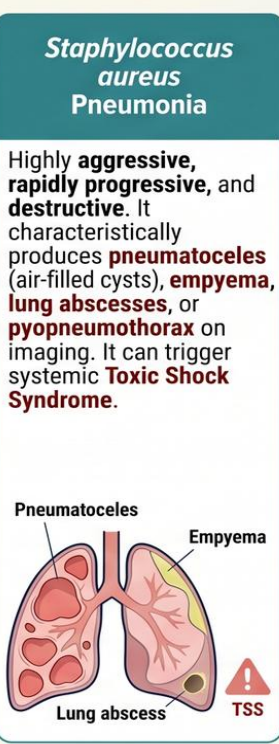
Appendix

SECTION 2: HIGH-YIELD CLINICAL CLUES FOR SPECIFIC PATHOGENS

While distinguishing pathogens clinically is difficult, certain features strongly point to specific organisms:

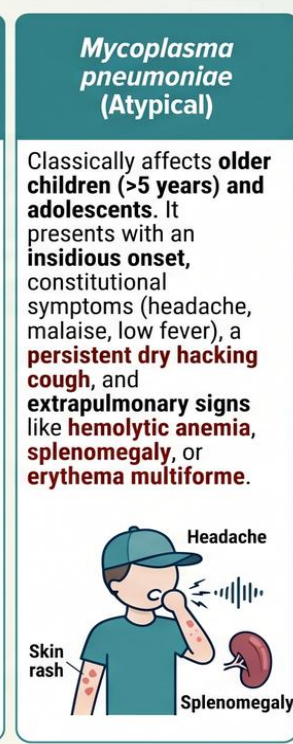
Staphylococcus aureus Pneumonia

Highly **aggressive, rapidly progressive**, and **destructive**. It characteristically produces **pneumatoceles** (air-filled cysts), **empyema**, **lung abscesses**, or **pyopneumothorax** on imaging. It can trigger systemic **Toxic Shock Syndrome**.



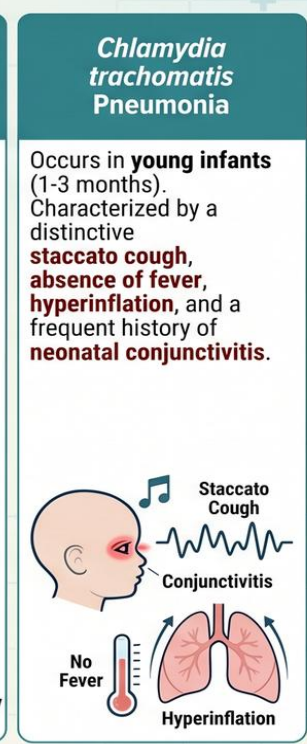
Mycoplasma pneumoniae (Atypical)

Classically affects **older children (>5 years) and adolescents**. It presents with an **insidious onset**, constitutional symptoms (headache, malaise, low fever), a **persistent dry hacking cough**, and **extrapulmonary signs** like **hemolytic anemia, splenomegaly, or erythema multiforme**.



Chlamydia trachomatis Pneumonia

Occurs in **young infants (1-3 months)**. Characterized by a distinctive **staccato cough**, **absence of fever**, **hyperinflation**, and a frequent history of **neonatal conjunctivitis**.



CHILDHOOD PNEUMONIA: ADVANCED DIAGNOSTICS & HOSPITALIZATION CRITERIA (PART 3)

✚ A HIGHLY ORGANIZED, CLEAN, AND SCANNABLE RESOURCE FOR CLINICIANS ✚

1. MULTIMODALITY STRUCTURAL IMAGING (FEATURES, PROS & CONS)

1. Chest X-ray

Shows **lobar, patchy, or interstitial infiltrates**.
Not needed for mild cases or if clinically improved.
Repeat if there is a poor treatment response, atelectasis, or round pneumonia.
Viral X-ray shows **hyperinflation** with **bilateral interstitial infiltrates** and **peribronchial cuffing**.
Radiographic appearance alone cannot determine exact etiology.



2. Thoracic Ultrasonography

Portable, low cost, highly accurate, and lacks ionizing radiation.
Its main disadvantage is being highly operator-dependent.

No Radiation



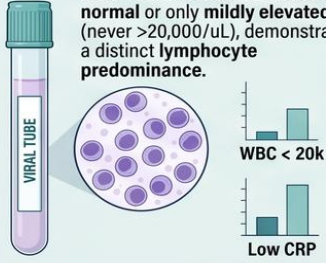
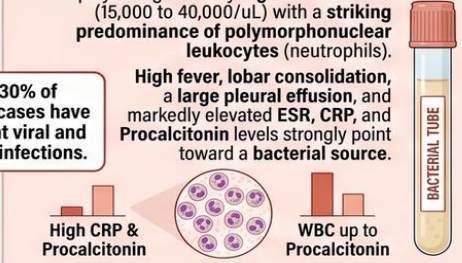
3. CT Scan

More accurate than X-ray, but has **higher cost, radiation, and requires sedation**.
Indicated to identify: pleural effusion, ground-glass opacities/halo signs (COVID-19), underlying congenital lung anomalies explaining poor response, or mediastinal masses.



2. LABORATORY DIAGNOSTIC SORTING (BACTERIAL vs. VIRAL)

Laboratory tracking helps differentiate the primary underlying inflammatory trigger:

Viral Pneumonia Pattern	Bacterial Pneumonia Pattern
WBC count is characteristically normal or only moderately elevated (never >20,000/uL), demonstrating a distinct lymphocyte predominance.  WBC < 20k Low CRP	Displays a significantly higher WBC count (15,000 to 40,000/uL) with a striking predominance of polymorphonuclear leukocytes (neutrophils). Note: 30% of pediatric cases have coexistent viral and bacterial infections. High fever, lobar consolidation, a large pleural effusion, and markedly elevated ESR, CRP, and Procalcitonin levels strongly point toward a bacterial source. High CRP & Procalcitonin  WBC up to 40k High CRP & Procalcitonin

3. ADVANCED MICROBIOLOGICAL ISOLATION METHODS

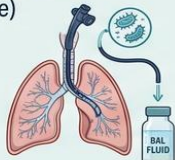
Definitive bacterial diagnosis requires **organism isolation from blood, pleural fluid, or lung tissue**.
Blood culture fails in most patients; sputum culture is of little value in young children.

Advanced pathways include:

1. Fiberoptic Bronchoscopy & BAL

(Bronchoalveolar Lavage)

Obtains **pure lower tract secretions** for culture; indicated for **persistent/recurrent pneumonia** or **atelectasis**, especially in immunocompromised hosts.



2. Percutaneous Lung Aspiration

Guided by **CT** for culture; carries a risk of **inducing air leaks** (pneumothorax), so reserved for **absolute treatment non-responders** or **immune-deficient patients**.



3. Targeted Testing

ASO/anti-DNase B for Group A Strep; **PCR** on nasopharyngeal specimens for Pertussis or respiratory viruses (RSV, influenza); **PCR** on **throat swab** (preferred) or **serology** (IgG/IgM) for Mycoplasma.



4. DEFINITIVE PEDIATRIC HOSPITALIZATION GRID

A child presenting with pneumonia must be **immediately admitted to the inpatient ward** if they meet **any** of the following **strict criteria**:

☒ **Age <6 months** or **immunocompromised state**.

☒ **Toxic appearance** or **clinical shock**.

☒ **Severe dehydration** or **inability to tolerate oral fluids/vomiting**.

☒ **Moderate to severe respiratory distress** (grunting, nasal flaring, chest retractions).

☒ **Hypoxemia** (Oxygen saturation SpO2 <90% on room air).

☒ **Complicated pneumonia** (pleural effusion, empyema, pneumatocele).

☒ **Sickle cell anemia** presenting with **acute chest syndrome**.

☒ **Failure to respond** to appropriate oral antibiotics or **high-risk pathogen suspicion**.

☒ **Social factors** (caregivers unable to safely provide home care or follow-up).



CHILDHOOD PNEUMONIA: TREATMENT PROTOCOLS & UNRESOLVED PNEUMONIA APPROACH (PART 4)

1. Outpatient vs. Inpatient Empirical Antibiotic Regimens

Outpatient Treatment (Mild cases)

Parameters:

Drug of choice: High-dose oral **Amoxicillin** at **90 mg/kg/24 hr** divided twice daily to overcome pneumococcal penicillin resistance.

Alternative: Oral **Cefuroxime**.

For school-age or suspected atypical infection (Mycoplasma/Chlamydia): Give oral **Azithromycin**; alternatives are **Clarithromycin** or **Doxycycline** (for ≥ 8 years).

Duration: Usually **5-7 days** for mild outpatient CAP; Azithromycin course is 5 days.



5-7 days

Inpatient Treatment (Hospitalized children)

Parameters:

Primary: Injectable **IV Ceftriaxone** (100 mg/kg/day) or **IV Cefotaxime**.

Atypical coverage: Add **IV Azithromycin** immediately if atypicals are suspected.

Staphylococcal coverage: Add **IV Vancomycin** or **Clindamycin** immediately if staphylococcal pneumonia is suspected based on **pneumatocele or empyema** on imaging.

Alternative substitution: **Ampicillin** or **Penicillin G** can be substituted only if the child is not severely ill, fully immunized against Hib and *S. pneumoniae*, and in areas without high resistance.

Duration: Antibiotics are continued until the child is afebrile for 72 hours, usually a total of **10 days**.



2. Targeted Viral, Influenza & COVID-19 Management

1. Presumed Viral & Influenza Pneumonia

Protocol:

Presumed Viral: Reasonable to **withhold antibiotics** in mildly ill pre-school children without respiratory distress; initiate antibiotics promptly if clinical status deteriorates.

Influenza Treatment & Prophylaxis: Oral **Oseltamivir** (Tamiflu) for children ≥ 2 weeks of age, ideally initiated **within 2 days of symptom onset**. Ribavirin has little documented benefit.



48h Initiate Oseltamivir within 2 days

2. COVID-19 Pneumonia Protocol

Flow Chart:

Intravenous IV Remdesivir for children aged **12-17 years**.

Flow Chart:

Increasing Oxygen Needs / Mechanical Ventilation

Systemic Dexamethasone



If failure to respond in 24 hours

Add Tocilizumab (IL-6 inhibitor) (Ages 2-11 years)

Prophylactic Anticoagulation: Indicated for all children ≥ 12 years.

3. Emergency Supportive Care Lines

Primary Measures Reduce Mortality:

1. **Supplemental Oxygen** via nasal cannula or mask to relieve hypoxemia, transitioning to CPAP or mechanical ventilation for impending respiratory failure.



2. **Vasoactive medications (adrenaline)** for concurrent septic shock.



3. **Oral Zinc Supplementation** administered for **7 days** has been proven to successfully reduce mortality among children with severe pneumonia in low- and middle-income countries.



7 Days: Reduces Mortality

1. Definition & Rule

Key Principle:

Persistence of symptoms or radiographic abnormalities beyond the expected time course.

While uncomplicated bacterial CAP improves clinically within **48-72 hours**, radiological abnormalities on X-ray can **normally persist for up to 3-6 months**.

Avoid unnecessary, prolonged antibiotic courses based solely on persistent X-ray abnormalities if the child is clinically well.



48-72 Hours (Clinical Recovery)



3-6 Months (X-ray Clearance)

Do Not Treat the X-ray!

4. The Unresolved Pneumonia Investigative Algorithm

2. Delayed Resolution Etiology

Causes:

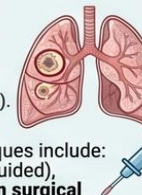
- Inadequate therapy, resistant organisms**, hidden immunodeficiencies.
- Nonbacterial causes** (fungi, TB), missed foreign body aspiration.
- Cystic fibrosis**, or congenital lung anomalies (sequestration).

3. Advanced Investigations

Procedures:

Repeat chest X-ray, **High-Resolution CT (HRCT)**, **ANCA** (for Wegener granulomatosis).

- Fiberoptic bronchoscopy with BAL**
- Definitive lung biopsy.** Biopsy techniques include: percutaneous transthoracic (CT/US-guided), transbronchial, thoracoscopic, or **open surgical biopsy**, reserved for critically ill, ventilated children without a diagnosis.



PNEUMONIA: UNDERSTANDING THE CONDITION

Definitions & Classifications

DEFINITIONS

Pneumonia: Inflammation of lung parenchyma, classified into lobar, lobular, or interstitial types^[cite: 1399, 1403].

Community-Acquired Pneumonia (CAP):

Signs/symptoms in a previously healthy child due to infection outside a hospital; leading cause of death worldwide in children <5 years^[cite: 1404, 1405].

Hospital-Acquired Pneumonia (HAP/Nosocomial):

Presents clinically after two days of hospital admission^[cite: 1406].

Ventilator-Associated Pneumonia (VAP):

Type of HAP presenting >48 hours after endotracheal intubation^[cite: 1407].

ETIOLOGY

1) Bacterial:

- **Gram-positive** (*S. pneumoniae*, *S. aureus*, Group A & B strep)^[cite: 1409, 1410];
- **Gram-negative** (*E. coli*, *H. influenzae*, *K. pneumoniae*)^[cite: 1411, 1412, 1413];
- **Atypical** (*M. pneumoniae*, *C. pneumoniae*)^[cite: 1414];
- *Mycobacterium tuberculosis*^[cite: 1415].

2) Viral:

RSV, adenovirus, influenza virus, SARS-CoV-2^[cite: 1416].

3) Fungal:

Aspergillus species, *Pneumocystis jirovecii*^[cite: 1417].

4) Parasitic:

Ascaris, *Strongyloides* species^[cite: 1418].

5) Non-infectious:

Drugs/irradiation, chemical pneumonitis, aspiration, hypostatic, hypersensitivity^[cite: 1419, 1423, 1424, 1425, 1426, 1428].

Löffler Syndrome: Pulmonary eosinophilic infiltrate due to parasites/drugs, mild symptoms^[cite: 1421, 1422].

SPECIFIC PATHOGEN FEATURES

S. aureus (pneumatoceles, empyema, pyopneumothorax); *S. pneumoniae* (consolidation, empyema); **Group B strep** (neonates); **Group A strep** (empyema); *Mycoplasma pneumoniae* (school-aged); *Chlamydia pneumoniae* (adolescents); *Chlamydia trachomatis* (infants); *Anaerobes* (aspiration in CP); *Hib* (unimmunized); *Yersinia pestis* (rat contact); *Aspergillus/P. jirovecii* (immunocompromised)^[cite: 1553-1545].

CLINICAL PICTURE

Symptoms: Frequently preceded by URI (rhinitis, cough)^[cite: 1430].

Acute onset: **Fever (high grade), Cough, Chest Pain**^[cite: 1431].

Tachypnea: Most consistent, sensitive, specific sign^[cite: 1432].

• WHO Fast Breathing Definitions:

Birth-2 mo (>60/min); 2 mo-1 yr (>50/min);
1-5 yrs (>40/min)^[cite: 1434, 1435, 1435].

Respiratory Distress: Grunting, nasal flaring, retractions, cyanosis^[cite: 1437, 1438].

Other Signs: Drowsiness, restlessness, anxiety, delirium^[cite: 1439].

Physical Exam: May be misleading in infants (only tachypnea)^[cite: 1440].

Local exam: Diminished breath sounds, crackles, rhonchi, bronchial breathing, pleural rub, increased fremitus, dullness^[cite: 1443-1449].

DIAGNOSIS

1) Chest X-ray: Shows lobar, patchy, interstitial infiltrates, complications^[cite: 1463, 1464, 1465]. **Not for mild/improving cases**^[cite: 1466, 1467]. Viral shows hyperinflation, interstitial infiltrates, peribronchovascular cuffing^[cite: 1469].

2) Ultrasonography: Low cost, accurate, portable, no radiation, operator-dependent^[cite: 1473, 1474, 1475].

3) CT Scan: Highly accurate (complications, anomalies, COVID-19), costly, radiation risk^[cite: 1476, 1477, 1479, 1481, 1482, 1483].

Etiology Isolation:

Bacterial: blood, sputum, BAL, lung aspiration (risky)^[cite: 1486-1495], Group A strep (ASOT/anti-DNase B)^[cite: 1496].

Pertussis, Mycoplasma, Viruses: PCR^[cite: 1497, 1499, 1502].

VIRAL VS. BACTERIAL

Viral: WBC normal/elevated (not >20,000/ml), lymphocyte predominance^[cite: 1548].

Bacterial: Higher WBC (15,000–40,000/ml), PMN predominance, large effusion, lobar consolidation, high fever, **higher ESR, CRP, procalcitonin**^[cite: 1549, 1550, 1551].

TREATMENT & MANAGEMENT

Hospitalization Indications: Age <6 months, immunocompromised, toxic, moderate/severe distress, hypoxemia (<90%), shock, severe dehydration, complicated, sickle cell with acute chest, vomiting, no oral response^[cite: 1558-1568, 1571].

Outpatient Treatment:

High-dose Amoxicillin (90 mg/kg/24 hr) or cefuroxime^[cite: 1574, 1575], **Azithromycin** for school-aged/atypical^[cite: 1576, 1577, 1578].

Inpatient Treatment: **Ceftriaxone (100 mg/kg/d)** or cefotaxime IV; add azithromycin (atypical); add vancomycin/clindamycin (staphylococcal)^[cite: 1579-1582].

Ampicillin/penicillin G for mild, immunized^[cite: 1583, 1584, 1585].

SPECIFIC TREATMENTS & SUPPORT

Viral Treatment: **Oseltamivir (Tamiflu)** for influenza^[cite: 1588, 1589]. Ribavirin for bronchiolitis^[cite: 1590]. **Antibiotics withheld in mild, presumed viral**^[cite: 1591, 1593, 1594, 1595].

COVID-19 Treatment: Remdesivir, Dexamethasone, Tocilizumab, Prophylactic anticoagulation^[cite: 1601-1604].

Treatment Duration: Until afebrile for 72 hrs (usually 10 days); shorter for mild (5-7 days); Azithromycin (5 days), Clarithromycin (7 days)^[cite: 1605-1608].

Supportive Care: **Supplemental oxygen**, CPAP, ventilation, vasoactive meds, **oral zinc (7 days)**^[cite: 1609-1614].

COMPLICATIONS & UNRESOLVED PNEUMONIA

Complications: Direct thoracic spread (effusion, empyema, abscess) or hematogenous spread (bacteremia, meningitis, endocarditis, arthritis, osteomyelitis)^[cite: 1552-1556].

Unresolved Pneumonia: Persistence > expected course^[cite: 1616, 1618]. Improvement in 48-72 hrs; radiological persistence for 3-6 months (avoid prolonged antibiotics)^[cite: 1552-1621].

Causes of Delayed Resolution: Inadequate therapy, resistant organisms, impaired host, nonbacterial, foreign body, cystic fibrosis, congenital anomalies, non-infectious diseases^[cite: 1622-1633].

Investigations: Repeat X-ray, HRCT, ANCA, bronchoscopy/BAL, lung biopsy^[cite: 1633].

CHAPTER 02

II. Neonatology



PHYSIOLOGICAL & PATHOLOGICAL JAUNDICE • PATHOLOGICAL JAUNDICE DETAILS • NEONATAL JAUNDICE OVERVIEW • COMPLICATIONS OF INDIRECT HYPERBILIRUBINEMIA • NEONATAL SEPSIS • PREMATURITY AND ITS COMPLICATIONS • TRANSIENT CUTANEOUS LESIONS

Understanding Neonatal Jaundice:

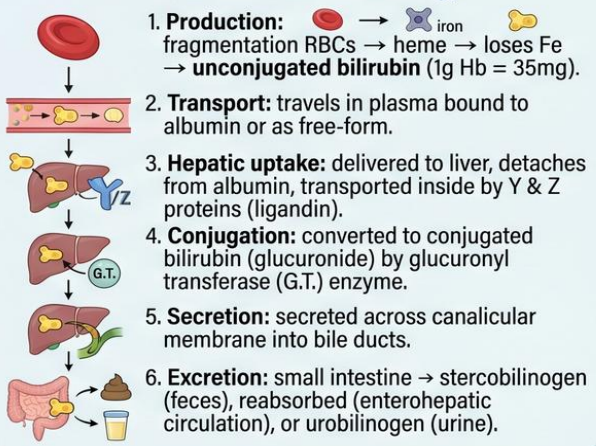
Physiological vs. Pathological (Differentiation & Physiology)

INCIDENCE & PHYSIOLOGY



Incidence: Most common problem. **60%** Full-term, **80%** Pre-term infants visibly jaundiced first week of life.

BILIRUBIN METABOLISM



NEONATAL JAUNDICE DEFINITION



Yellowish discoloration of skin, mucus membranes, body fluids due to bilirubin increase > normal level (1 mg/dl).

PHYSIOLOGICAL JAUNDICE (CRITERIA, CAUSES, RISK FACTORS)

CRITERIA OF PHYSIOLOGIC JAUNDICE



Day 2-3
7d



FULL-TERM: appears 2nd–3rd day, peak <12 mg/dl, disappears <7 days, asymptomatic, no treatment.



Day 4-6
10-14d



PRETERM: appears 4th–6th day, peak <14 mg/dl, disappears 10–14 days, asymptomatic, no treatment. Slower rise, longer duration, higher levels due to immature excretion.

CAUSES OF PHYSIOLOGIC JAUNDICE



Increased bilirubin production (fetal RBC breakdown).



Transient limitation in hepatic uptake & conjugation.



Increased enterohepatic circulation.

RISK FACTORS FOR EXAGGERATED PHYSIOLOGIC JAUNDICE

- 

Maternal diabetes, **Race** (East Asian, Native American),
- 

Prematurity, Polycythemia, Male sex, Trisomy 21,
- 

Cutaneous bruising, Cephalohematoma,
- 

Oxytocin induction, Drugs (Vitamin K3, Novobiocin),
- 

Weight loss (dehydration/caloric deprivation),
- 

Delayed bowel movement, Sibling with physiologic jaundice.

PATHOLOGIC JAUNDICE (DIFFERENTIATION CRITERIA)

CRITERIA TO SUSPECT PATHOLOGIC JAUNDICE ⚠

Suspect if ANY are present:

- 

<24 hrs
- 

↑
- 

>12/>14
- 

>1 week/
>2 weeks
- 

>20%
- 

Family history of hemolytic disease.
- 

Clinical presence of pallor, hepatomegaly, or splenomegaly.
- 

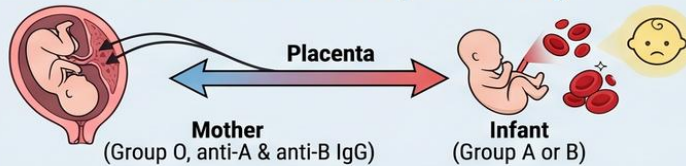
Failure of **phototherapy** to lower bilirubin level.
- 

Baby **clinically ill** (vomiting, lethargy, poor feeding, weight loss, apnea, bradycardia, hypothermia, light stools, dark urine).
- 

Signs of **kernicterus** are present.
- 
 1. Appears in **first 24 hours** of life.
 2. Serum bilirubin rising **>5 mg/dL/24 hr.**
 3. Serum bilirubin **>12 mg/dL** (full-term) or **>14 mg/dL** (preterm).
 4. Jaundice persists >1 week (full-term) and >2 weeks (preterm).
 5. Direct-reacting bilirubin **>20%** of total bilirubin at any time.
 6. Family history of hemolytic disease.
 7. Clinical presence of pallor, hepatomegaly, or splenomegaly.
 8. Failure of **phototherapy** to lower bilirubin level.
 9. Baby **clinically ill** (vomiting, lethargy, poor feeding, weight loss, apnea, bradycardia, hypothermia, light stools, dark urine).
 10. Signs of **kernicterus** are present.

Pathological Jaundice: Causes & Key Conditions

1. ABO Incompatibility



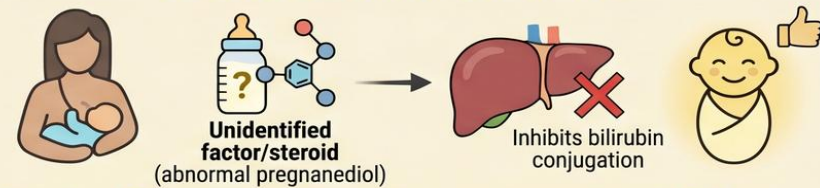
- 💡 **Most common hemolytic disease** (20-25% pregnancies, 10% develop disease due to low antigenicity). **Mother has O blood**, Infant has A or B. Anti-A/B IgG **cross placenta**.
- ➡ Clinical manifestations **usually mild**; jaundice only sign (**1st 24 hours**). Pallor absent, hydrops fetalis extremely rare, liver/spleen not greatly enlarged. **Rare severe cases** lead to kernicterus.
- ➡ **Lab**: ABO incompatibility, weakly to moderately positive **direct Coombs test**, **Hyperbilirubinemia**, Hemoglobin usually normal (can drop to 10-12 g/dL), **Reticulocytes increased** (10-15%). Unconjugated bilirubin may reach >20 mg/dL (10-20% cases).
- 💡 **Treatment**: Phototherapy, rare exchange transfusions, potential packed RBC transfusion.

3. Galactosemia



- ➡ **Inborn error of metabolism** causing **liver cell injury** and **toxic damage**.
- ➡ Presents as pathological jaundice (after 3rd day, within 1st week, or **persistent**). Direct, indirect, or biphasic **hyperbilirubinemia**. Predisposes infant to **E. coli sepsis**.

2. Breast Milk Jaundice



- ➡ **Conjugation defect** caused by unidentified factor/steroid in milk **inhibiting conjugation**. Presents **second week** (occasionally earlier), regresses within 3 months.
- ➡ Diagnosed **by exclusion**. Infant remains **entirely well and thrives**. **Not a contraindication** to breastfeeding (temporary 24-48hr cessation considered for dx/rx).
- *Distinguish from early-onset (2nd-3rd day due to dehydration/caloric deprivation).

4. Extrahepatic Biliary Atresia



- ➡ **Excretion defect**; **most common neonatal cholestasis** (85%). Prolonged elevation of **conjugated bilirubin** (>14 days). Jaundice persisting **>2 weeks**, **acholic stools**, **dark urine**. Complete obliteration of extrahepatic biliary tree at/above porta hepatis (difficult surgical problem).
- ➡ **Diagnostic evaluation**: direct bilirubin, LFTs, sonography, alpha-1-antitrypsin, sweat test, biopsy, laparotomy, direct cholangiography.
- ➡ **Management**: direct surgical drainage (correctable) or **Kasai procedure** (uncorrectable).

Note: This is an explanatory graphic based on clinical descriptions. Consult a medical professional for diagnosis and treatment.

DIAGNOSIS OF NEONATAL JAUNDICE

DEFINITION & OVERVIEW



Neonatal jaundice: **YELLOW** discoloration of SKIN, SCLERA, and MUCOUS MEMBRANES from **ELEVATED SERUM BILIRUBIN**.

Common condition, can be **PHYSIOLOGICAL** or **PATHOLOGICAL**.



Cause: Increased **RED BLOOD CELL** turnover + immature **HEPATIC CONJUGATION**.

EARLY RECOGNITION OF PATHOLOGICAL JAUNDICE IS ESSENTIAL to prevent **KERNICTERUS & NEUROLOGICAL DAMAGE**.



PHYSIOLOGICAL JAUNDICE

(Commonest type)

Appears **AFTER 24 HOURS**, usually day 2 or 3. Due to fetal RBC destruction & immature liver.



CLINICAL FEATURES:



Mild jaundice in **HEALTHY** infant



Peak bilirubin at day 3-5



Baby **FEEDS WELL**, **ACTIVE**



NO hepatosplenomegaly or anemia



NORMAL urine & stool color



NORMAL urine & stool color



COURSE: RESOLVES within 7 days (term), up to 14 days (preterm). **NO TREATMENT**, observation & adequate feeding



PATHOLOGICAL JAUNDICE

(Caused by **UNDERLYING DISEASE**)



REQUIRES PROMPT EVALUATION to prevent bilirubin encephalopathy/kernicterus.



SUSPECTED WHEN:



Jaundice **WITHIN FIRST 24 HOURS**

- **RAPID** bilirubin rise (>5 mg/dL/day)
- Bilirubin **EXCEEDS** expected physiological levels



PERSISTS beyond 14 days (term)

- Infant is **ILL**, **LETHARGIC**, **POOR FEEDING**



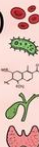
CONJUGATED hyperbilirubinemia

- **PALE STOOLS** or **DARK URINE**



COMMON CAUSES:

- Hemolytic diseases (ABO/Rh incompatibility)
- Infections (e.g., neonatal sepsis)
- Metabolic disorders (e.g., galactosaemia)
- Hepatobiliary diseases (e.g., biliary atresia)
- Endocrine disorders (e.g., congenital hypothyroidism)



DIAGNOSIS DEPENDS ON:

- **Age at onset**
- **Duration & Severity**
- **Feeding history**
- **Stool & Urine color**
- **Presence of Anemia, Hepatomegaly, Illness**
- **LABORATORY INVESTIGATIONS:** Serum Bilirubin & Blood Grouping



DIFFERENCE: PHYSIOLOGICAL vs. PATHOLOGICAL JAUNDICE

FEATURE	PHYSIOLOGICAL	PATHOLOGICAL
1. TIME OF APPEARANCE	2nd-3rd day (Term), 4th-6th day (Preterm)	WITHIN 1ST 24 HOURS
2. LEVEL OF BILIRUBIN	<12 mg/dl (Term), <14 mg/dl (Preterm)	>12 mg/dl (Term), >14 mg/dl (Preterm)
3. TYPE OF BILIRUBIN	USUALLY UNCONJUGATED	UNCONJUGATED or CONJUGATED
4. RATE OF RISE	SLOWER than 5mg/dl/24hrs	FASTER than 5mg/dl/24hrs
5. SEVERITY	MILD	OFTEN SEVERE
6. TIME OF DISAPPEARANCE	Within 7 days (Term), 10-14 days (Preterm)	PERSISTENCE >1 week (Term), >2 weeks (Preterm)
7. CLINICAL CONDITION	ASYMPTOMATIC	BABY MAY BE SICK
8. URINE & STOOL COLOR	NORMAL COLOR	PALE STOOL , DARK URINE
9. TREATMENT	OBSERVATION ONLY	INVESTIGATION & TREATMENT

CORE CONDITIONS OF PATHOLOGICAL JAUNDICE

1. BREAST MILK JAUNDICE

Occurs in healthy breastfed infants **AFTER 1st week**. Substances in breast milk increase bilirubin enterohepatic circulation.

- Appears 1st-2nd week
- Mild prolonged unconjugated jaundice
- Infant **HEALTHY**, **GAINING WEIGHT**
- **NORMAL** urine & stool

Breastfeeding usually **CONTINUES**.



2. ABO INCOMPATIBILITY

Hemolytic cause from maternal antibodies against fetal RBCs.

- **EARLY** jaundice within first 24 hours
- **ANEMIA** & **HEMOLYSIS**
- **ELEVATED** unconjugated bilirubin
- **POSITIVE** Coombs test

Treatment: **PHOTOTHERAPY** or **EXCHANGE TRANSFUSION**.



3. GALACTOSAEMIA

INHERITED metabolic disorder, deficiency of galactose metabolism.

- **SEVERE** jaundice after milk feeding
- **VOMITING**, **POOR FEEDING**
- **HEPATOMEGALY**
- **HYPOGLYCEMIA**
- Reducing substances in urine
- **RISK OF SEPSIS**

Treatment: **LACTOSE-FREE & GALACTOSE-FREE DIET**.



4. EXTRAHEPATIC BILIARY ATRESIA

CONGENITAL obstruction of bile ducts.

- **CONJUGATED HYPERBILIRUBINEMIA**
- **PALE STOOLS**
- **DARK URINE**
- **HEPATOMEGALY**
- **FAILURE TO THRIVE**

Treatment: **SURGICAL INTERVENTION** (Kasai procedure).



MAJOR COMPLICATIONS

1. INSPISSATED BILE SYNDROME

KEYPOINTS: **PERSISTENT ICTERUS**ASSOCIATION: Significant elevations in **BOTH DIRECT & INDIRECT BILIRUBIN** in infants with **HEMOLYTIC DISEASE**

CAUSE UNCLEAR

OUTCOME: Jaundice **CLEARs SPONTANEOUSLY** within weeks/months.

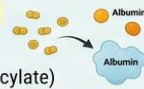
2. KERNICTERUS (BILIRUBIN ENCEPHALOPATHY)

KEYPOINT: **NEUROLOGIC SYNDROME**FROM **DEPOSITION** of **UNCONJUGATED BILIRUBIN** IN **BASAL GANGLIA & BRAINSTEM NUCLEI**

PREDISPOSING FACTORS

UNCONJUGATED HYPERBILIRUBINEMIA**INCREASED FREE BILIRUBIN**

- **HYPOALBUMINEMIA**
- **COMPETITIVE BINDING** (e.g., Vitamin K3, Sulpha, Salicylate)



CLINICAL MANIFESTATIONS

Usually 2-5 days (Term) or up to 7th day (Preterm)

RISK ZONE: Bilirubin > 30 mg/dL (Range 21-50 mg/dL)

ACUTE FORM PHASES



PHASE 1 (1st 1-2 days):
POOR SUCKING, LOSS OF MORO REFLEX, STUPOR, HIGH-PITCHED CRY, HYPOTONIA, SEIZURES.



PHASE 2 (Middle 1st week):
HYPERTONIA, OPISTHOTONOS, RETROCOLLIS, BULGING FONTANEL, FEVER.



PHASE 3 (After 1st week):
HYPERTONIA. SERIOUS DAMAGE, DEATH, OR TEMPORARY RECOVERY.



INCREASED BLOOD-BRAIN BARRIER PERMEABILITY

- **SEPSIS**
- **HYPOXIA**
- **ACIDOSIS**



CHRONIC FORM

LATER 1ST YEAR: Recurrent OPISTHOTONOS, RIGIDITY, IRREGULAR MOVEMENTS, CONVULSIONS.**2ND YEAR:** Decreased Opisthotonos, Increased INVOLUNTARY MOVEMENTS, RIGIDITY/HYPOTONIA.

BY 3 YEARS: COMPLETE SYNDROME: BILATERAL CHOREOATHETOSIS, EXTRAPYRAMIDAL SIGNS, SEIZURES, MENTAL DEFICIENCY, DYSARTHRIC SPEECH, HIGH-FREQUENCY HEARING LOSS, SQUINTING, DEFECTIVE UPWARD EYE MOVEMENT.



⊗ PREVENTABLE CAUSES

Early discharge (<48 hr) with **NO FOLLOW-UP**
Failure to check bilirubin <24 hr
Failure to recognize **RISK FACTORS**
Visual UNDERESTIMATION
LACK OF CONCERN
DELAY in measuring/phototherapy
Failure to **RESPOND** TO PARENTAL CONCERN

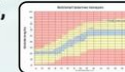
COMPLICATIONS OF INDIRECT HYPERBILIRUBINEMIA:
MAJOR RISKS & MANAGEMENT

INDICATIONS

BELOW EXCHANGE THRESHOLDS, e.g., <20 mg/dL (TERM), Weight-based (PRETERM).

METHODS

SPECIAL BLUE FLUORESCENT TUBES or **FIBEROPTIC BLANKETS**. EYES COVERED.



MANAGEMENT OF INDIRECT HYPERBILIRUBINEMIA

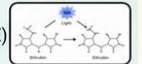
PHOTOTHERAPY

GOAL: CLINICAL JAUNDICE REDUCTION via **HIGH-INTENSITY LIGHT** (BLUE SPECTRUM: 420-470 nm).



MECHANISM

PHOTO-ISOMERIZATION: Converts **TOXIC UNCONJUGATED BILIRUBIN** into **CONFIGURATIONAL ISOMER** (EXCRETED IN BILE) & **STRUCTURAL ISOMER LUMIRUBIN** (EXCRETED BY KIDNEYS).



METHODS

SPECIAL BLUE FLUORESCENT TUBES or **FIBEROPTIC BLANKETS**. EYES COVERED.



COMPLICATIONS ⚠

LOOSE STOOLS
ERYTHEMATOUS RASH
ERYTHEMATOUS RASH
OVERHEATING & DEHYDRATION

CHILLING
BRONZE BABY SYNDROME
EYE INJURY/NASAL OCCLUSION
DNA/RETINAL DAMAGES

EXCHANGE TRANSFUSION

GOALS

REMOVE UNCONJUGATED BILIRUBIN, **REMOVE ANTIBODIES**, **RESTORE BLOOD VOLUME**, **CORRECT ANEMIA**, **PROVIDE FREE ALBUMIN**, **REPLACE SENSITIZED RBCs**.



METHOD

EMPTY STOMACH, **UMBILICAL VEIN CANNULATION** (Max 7 cm), **FRESH BLOOD** (Type O Rh-neg or matching), **TWO BLOOD VOLUMES** (2 x 85 mL/kg) at 35-37°C, 45-60 min, **ALTERNATING 20 mL ASPIRATION/INFUSION**. 1cc **CALCIUM GLUCONATE IV** after each 100cc.



MONITOR BILIRUBIN every 4-8 hr for **REBOUND**.

INDICATIONS

IMMEDIATE: Cord Hb <=8 g/dL or **HYDROPS**

EARLY: Cord Hb <=10 g/dL, Cord Bilirubin >=5 mg/dL, Bilirubin Rise >0.5-1.0 mg/dL/hr, Previous Kernicterus, Severe Sibling Erythroblastosis, Reticulocytes >15%, Prematurity, Indirect Bilirubin >20 mg/dL with Hemolysis.



COMPLICATIONS ⚠

TRANSIENT BRADYCARDIA
CYANOSIS
ELECTROLYTE DISTURBANCES
THROMBOSIS
APNEA
DEATH

INFECTIOUS RISKS (CMV, HIV, Hepatitis)
NECROTIZING ENTEROCOLITIS
LATE ANEMIA
GRAFT VS HOST REACTION
INSPISSATED BILE
PORTAL VEIN THROMBOSIS

OTHER THERAPIES



PHENOBARBITONE (Enzyme Inducer)

INTRAVENOUS IMMUNOGLOBULIN (IVIG) (for Coombs-positive hemolytic anemia)

TIN-PROTOPORPHYRIN (Heme Oxygenase Inhibitor).



NEONATAL SEPSIS & CLINICAL SEPSIS SCORE

DEFINITION: Neonatal sepsis is a clinical syndrome of systemic illness resulting from the pathophysiologic effects of local or systemic infection accompanied by bacteremia in the first 4 weeks of life. In 30% of cases it is associated with meningitis, and in 20% with pneumonia or urinary tract infection.



RISK FACTORS

1. MATERNAL:

Premature rupture of membranes (>24 hr), peripartum fever ($\geq 38.0^{\circ}\text{C}$) or proved infection, chorioamnionitis, urinary tract infection, perineal *E. coli* or vaginal GBS colonization, meconium-stained or foul-smell amniotic fluid, and maternal diabetes or pre-eclampsia.



2. NEONATAL:

****Prematurity** (the single most important risk factor), multiple gestations, galactosemia (predisposed to *E. coli* sepsis), immune defects, asplenia, and birth depression requiring resuscitation.



3. ENVIRONMENTAL/NOSOCOMIAL:

Crowding, inadequate nursing staff, lack of hand washing, invasive procedures, prolonged antibiotic use, and exposure to adult infection.



CAUSATIVE ORGANISMS: *E. coli*, *Klebsiella*, *Enterococci*, *Proteus*, *Pseudomonas*, *Listeria monocytogenes*, and Group B streptococci (GBS).



CLINICAL PATTERNS

1. EARLY ONSET SEPSIS: Present at birth or during the first week (usually <3 days); a fulminant multisystem illness acquired during the intrapartum period from the maternal genital tract or transplacentally; history of obstetric complications is common.



2. LATE ONSET SEPSIS: Appears 8-28 days after delivery; organism is acquired from the genital tract or environment; can be multisystem or focal; obstetric complications are uncommon.



3. NOSOCOMIAL SEPSIS: Appears after about 8 days from hospital admission; related to the underlying illness and the flora of the NICU; can be multisystem or focal.



CLINICAL PICTURE



EARLY SIGNS: Subtle ("baby is not doing well", poor feeding, fever or hypothermia, edema).



GASTROINTESTINAL: Abdominal distention, vomiting, diarrhea, and hepatomegaly.



RESPIRATORY: Tachypnea, retractions, nasal flaring, grunting, apnea, and cyanosis.



RENAL: Oliguria (prerenal failure).



CARDIOVASCULAR: Pallor, mottling, cold clammy skin, tachycardia, bradycardia, and hypotension.



CNS: Lethargy, irritability, tremors, seizures, hypotonia, hyporeflexia, poor Moro reflex, full fontanelle, and high-pitched cry.



HEMATOLOGIC: Pallor, jaundice, splenomegaly, petechiae, purpura, and bleeding/DIC.

CLINICAL SEPSIS SCORE:

Sepsis is strongly suspected if a baby exhibits **3 or more** of the following signs (each scores 1 point):



1. Apnea, tachypnea, cyanosis, respiratory distress.
2. Bradycardia, tachycardia.
3. Hypotonia, seizure.
4. Poor skin color, poor peripheral circulation.
5. Irritability, lethargy, poor feeding.



6. Hepatosplenomegaly, abdominal distention.
7. Fever, hypothermia.

DIAGNOSTIC TYPES:

POSSIBLE SEPSIS: (Presence of clinical signs);



PROBABLE SEPSIS: (Suggestive clinical picture plus one or more criteria: risk factors, positive septic screen, or radiological pneumonia);



PROVEN SEPSIS: (Clinical sepsis plus a positive culture from blood, CSF, or urine).



INVESTIGATIONS:



I. EVIDENCE OF INFECTION: Culture from normally sterile sites (blood, CSF, urine via catheter/suprapubic puncture; urine culture not necessary <24 h unless indicated), focus abscess culture, or Gram stain demonstration.



III. EVIDENCE OF MULTIORGAN DISEASE: Metabolic acidosis (pH, Pco_2), pulmonary dysfunction, renal tests (BUN, creatinine), and hepatic injury tests.



II. EVIDENCE OF INFLAMMATION: CBC showing neutrophilia or neutropenia (severe sepsis sign), band to total neutrophil ratio >0.2 , toxic granulation, thrombocytopenia (late finding), fragmented RBCs, increased



IV. RADIOLOGICAL STUDIES: Chest X-ray or urinary tract imaging.



TREATMENT & PREVENTION:

TREATMENT:



SUPPORTIVE MEASURES: Warming, fresh frozen plasma for DIC, fresh blood for anemia/antibodies, fluid/electrolyte correction, ventilatory support, shock correction (fluids and dopamine 2-20 mcg/kg/min), and treating convulsions/acidosis.



SPECIFIC TREATMENT: Immediate empiric antibiotic therapy (nosocomial coverage typically starts with vancomycin plus a third-generation cephalosporin, modified after culture results). Antibiotic duration is 2 weeks for sepsis and 3 weeks for meningitis.



OTHER LINES: IV immunoglobulins, GM-CSF, or exchange transfusion.

PREVENTION:

MEASURES: Strict hand washing, cleaning incubators, universal precautions (gowns/gloves), limiting nurse-to-patient ratios, avoiding crowding, early enteral feeding, and isolating infectious babies.





PREMATURITY & ITS COMPLICATIONS:

Definitions, Causes, Handicaps, Management & Outcomes



DEFINITIONS



Live-born < 37 completed weeks (WHO).

Low Birth Weight (LBW) ≤ 2500 g (due to prematurity, IUGR/SGA, or both). Associated with increased morbidity & mortality.

Incidence: 6-7% of births or more.

CAUSES



1. **Idiopathic** (Most common).



2. **Maternal:** Age (<20 or >40 yrs), Pregnancy-related (pre-eclampsia, abruption, placenta previa, chorioamnionitis, incompetent os, polyhydramnios, iatrogenic, PROM), Non-pregnancy related (trauma, chronic illness - DM, heart/renal disease, infections, drug abuse).



3. **Fetal:** Distress, anomalies, multiple gestation, erythroblastosis, nonimmune hydrops.



4. **Low Socioeconomic Standard.**

COMPLICATIONS



1. **Respiratory:** RDS (HMD), BPD, pneumothorax, apnea, congenital pneumonia, pulmonary hemorrhage.



2. **Cardiovascular:** PDA, hypotension, hypertension, bradycardia.



3. **Hematologic:** Anemia, hyperbilirubinemia, hemorrhage, DIC, Vitamin K deficiency.



4. **Gastrointestinal:** Feeding difficulty, NEC, perforation.



5. **Metabolic-Endocrine:** Hypocalcemia, hypoglycemia.



5. **Metabolic-Endocrine:** Hypocalcemia, hypoglycemia.



6. **CNS:** IVH, PVL, HIE, seizures, ROP, deafness, kernicterus.



7. **Renal:** Electrolyte disturbances, acidosis, glycosuria, edema.



8. **Immunologic:** Infection, sepsis.



9. **Nutritional:** Rickets, nutritional disorders.



10. **Management Complications:** ROP, air leak syndrome, sepsis, BPD, hypothermia.

PHYSIOLOGICAL HANDICAPS

(Immature Systems)



1. **Thermoregulatory:** Immature center, large surface area, reduced fat/muscle activity.



2. **Respiratory:** Poor center/muscles, surfactant deficiency, underdeveloped alveoli, poor reflexes.



3. **Gastrointestinal:** Poor suck/swallow, weak sphincter, low capacity/acidify, decreased fat/vitamin absorption, hypotonic colon.



4. **Hepatic:** Impaired bilirubin conjugation/excretion, Vitamin K deficiency.



5. **Hemostatic:** Fragile capillaries, inadequate tissues, impaired clotting.



6. **Renal:** Poor urine concentration, low GFR/clearance.



7. **CNS:** Poorly developed brain, soft skull, immature blood-brain barrier.



8. **Blood:** Immature organs, poor iron stores, hemolysis.



9. **Immunological:** Reduced antibodies/phagocytosis/response.



10. **Nutritional:** Poor reflexes, decreased digest/absorption, low stores.

SURVIVAL RATES



<750g (25%)
750-1000g (55%)
1000-1500g (85%)
1500-2000g (95%)
2000-2500g (98%)

CLINICAL FEATURES



LBW, small measurements, weak, hypoactive, sleepy, feeble cry, poor suckling, relatively large head/abdomen, thin red shiny translucent skin, diminished fat, jerky respiration with apnea, underdeveloped ear cartilage, breast nodules, genitals, sole creases.

MANAGEMENT



Goals: Temp maintenance (36.5-37°C rectal), proper feeding, infection protection, deficiency prevention. Incubation for temp regulation (28-34°C), humidity (40-60%), oxygen, contamination reduction.



Feeding: Small-volume (breast milk/preterm formula) started 2-4 hrs if stable. **Methods:** Suck/swallow (breast/bottle), swallow only (dropper/spoon), neither (NG tube/IVF). Tolerability check. Specific amounts for <1000g, 1000-1500g, >1500g with daily increments. IVF for intolerability, distress, hemorrhage, NEC.



Infection Protection: Incubation, minimal handling, strict sterilization, hand washing.



Deficiency Prevention: Vit K (birth), Vit D, Vit A, Vit C, Folic acid, Iron (from 6th week).

SEQUELAE OF LBW



Hypoxia, ischemia, mental retardation, spastic diplegia, microcephaly, seizures, poor school performance, IVH, hydrocephalus, hearing/visual impairment, ROP, strabismus, myopia, BPD, cor pulmonale, bronchospasm, short-bowel syndrome, liver disease, cirrhosis, osteopenia, fractures, child abuse, SIDS.

PRETERM IMMUNIZATION



Full doses at chronological age. Oral polio contraindicated (give IPV). Acellular DPT if hospitalized. Pertussis contraindicated if documented brain instability.

TRANSIENT CUTANEOUS LESIONS: UNDER SKIN EVALUATIONS IN NEWBORNS

Common temporary skin conditions often seen in neonates, requiring understanding but rarely treatment.

1. VASOMOTOR INSTABILITY & PERIPHERAL CIRCULATORY SLUGGISHNESS



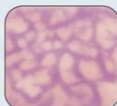
Acrocyanosis

Harmless cyanosis of hands & feet, especially when cool.



Mottling or Cutis Marmorata

General circulatory instability, lace-like pattern. May link to serious illness or transient temperature fluctuation.



Harlequin color change

Rare division of body from forehead to pubis into RED & RED & PALE halves when lying on side. Transient & harmless.



2. THE VERNIX CASEOSA

Whitish or clay-colored, CHEESY & GREASY material covering body; decreases near term, absent in post-term infants.



3. LANUGO HAIR

FINE, SOFT, immature hair frequently covering SCALP & BROW. May cover face of premature infants; usually replaced by vellus hair in full-term.



4. SALMON PATCH

Common transitory macular CAPILLARY HEMANGIOMAS located on eyelids and back of the neck.



5. SCATTERED PETECHIAE

Small red dots may be observed on presenting part (usually SCALP OR FACE) after a DIFFICULT DELIVERY.



6. MONGOLIAN SPOTS

Blue, well-demarcated PIGMENTATION over buttocks, back, & other parts in >50% of newborns. Fade within first year; require REASSURANCE.



7. ERYTHEMA TOXICUM

Small, white, occasionally VESICULOPUSTULAR PAPULES on an erythematous base. Develop 1-3 days after birth, persist ~1 week, contain eosinophils. Face, trunk, extremities.



8. PUSTULAR MELANOSIS

Benign lesion predominantly in BLACK NEONATES. Neutrophil-containing VESICULOPUSTULAR ERUPTION at birth around chin, neck, back, extremities, palms, or soles; lasts 2-3 days.



9. MILIAA OF THE FACE

Distended SEBACEOUS GLANDS producing tiny WHITE PIN-POINT PAPULES. Frequently present, disappear in 2-3 weeks. Mainly NOSE & CHIN.



10. PHYSIOLOGICAL JAUNDICE

Yellowish skin tint, may be present on the 2nd to 4th DAYS after birth.



11. AMNIOTIC BANDS

Fibrous bands may disrupt skin, causing EXTREMITY DEFECTS (amputation, ring constriction, syndactyly), FACIAL DEFECTS, or TRUNK DEFECTS. Cause uncertain.



CHAPTER 03

III. Renal Diseases



PEDIATRIC HEMATURIA EVALUATION • ACUTE NEPHRITIC SYNDROME & APSGN • NEPHROTIC SYNDROME • ACUTE KIDNEY INJURY • CHRONIC KIDNEY DISEASE • URINARY TRACT INFECTIONS

PEDIATRIC HEMATURIA: APPROACH & EVALUATION

1. Definition & Quantitative Screening

Core Definition: Hematuria is the abnormal presence of red blood cells (RBCs) in the urine.



Microscopic Hematuria



- Quantified as ≥ 3 RBCs per high-power field (HPF) in a centrifuged urine specimen on at least 2-3 consecutive samples.
- Usually asymptomatic, discovered incidentally.

Gross (Macroscopic) Hematuria



- Visually obvious macroscopic blood causing a red, brown, or smoky urine color.
- Requires immediate systematic diagnostic evaluation.

2. Anatomical & Etiological Source Localization



Glomerular Source (Upper Urinary Tract)

- Pathophysiology:** Glomerular filtration barrier injury.
- Urine Color:** Cola/tea/smoky colored, due to methemoglobin.
- Proteinuria:** Significant $> 2+$.
- RBC Morphology:** Distorted, dysmorphic RBCs (acanthocytes), blebs.
- Urinary Casts:** Presence of Red blood cell casts.



Extra-Glomerular Source (Lower Urinary Tract)

- Pathophysiology:** Structural/vascular lesions, trauma, infection.
- Urine Color:** Bright red/pink urine, often with clots.
- Proteinuria:** Absent or minimal $< 1+$.
- RBC Morphology:** Round, normal isomorphic RBCs.
- Urinary Casts:** Absence of urinary casts.



3. Diagnostic Approach & Practical Panel



1. Analysis of Fresh Urine:

Evaluate RBC shape via phase-contrast microscopy, crystals, bacteria.



2. Abdominal Ultrasound:

Screen for renal calculi, duplication, hydronephrosis, Wilms tumor.



3. Renal Function Tests:

Serum creatinine, BUN, electrolytes for acute renal insufficiency.



4. Blood Picture/CBC & Coagulation Profile:

Screen for bleeding diatheses, systemic issues.



5. Targeted Serological Screening:

ASO/Anti-DNase titers & low C3 for post-infectious GN; ANA/anti-dsDNA for Lupus Nephritis.



4. Key Indications for Renal Biopsy



1. Unexplained persistent or recurrent gross hematuria.



2. Biopsy-confirmed or highly suspected Lupus Nephritis.



3. Atypical Glomerulonephritis course: nephritic-nephrotic syndrome or absent/persistently low C3 level beyond 6-8 weeks.



4. Unexplained or progressive acute kidney injury/renal failure.

ACUTE POST-STREPTOCOCCAL GLOMERULONEPHRITIS (APSGN)



Definition

Acute specific self-limited glomerulonephritis triggered by prior **streptococcal infection** characterized by a sudden onset of **gross hematuria**, **mild to moderate edema**, **hypertension**, and potential **oliguria**.



Etiology and Epidemiology

Follows throat or skin infection by specific **nephritogenic strains** of group A Beta-hemolytic streptococci.



Comparison of AGN after Throat vs Skin Infection

Throat Infection

Type 4, 12 strains;
Winter/Spring
Latent period: **1-2 weeks**
Elevated ASO titers

Skin Infection

Type 49 strain;
Summer
Latent period: **2-3 weeks**
ASO titers may be normal

Clinical Manifestations



- Primarily affects ages 5-12 years. Typical presentation is **acute nephritic syndrome** 1-2 weeks after pharyngitis or 3-6 weeks after pyoderma.
- Edema** results from salt and water retention.
- Patients can develop **encephalopathy** and/or **heart failure** from hypertension or hypervolemia, or encephalopathy from toxic bacterial effects.
- Nonspecific symptoms: malaise, lethargy, abdominal/flank pain, fever.
- Acute phase resolves within 6-8 weeks; protein excretion and hypertension normalize by 4-6 weeks; **microscopic hematuria** can persist for 1-2 years.

Complications

Result from **acute hypertension** and **renal dysfunction**. Hypertension in 60%; **hypertensive encephalopathy** in Other complications: **heart failure**, **hyperphosphatemia**, **hypocalcemia**, **acidosis**, **seizures**, **uremia**.

Medical emergencies in APSGN include: **hyperkalemia**, **hypertensive encephalopathy**, **congestive heart failure**, and **rapidly progressive glomerulonephritis (RPGN)**.

Pathogenesis



Mediated by **immune complexes** as strongly indicated by structural evaluations and **depression in serum complement (C3) levels**.

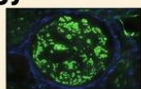
Pathology



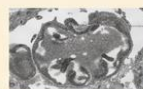
Symmetrically enlarged kidneys



Light microscopy: **Enlarged, relatively bloodless glomeruli** with diffuse mesangial cell proliferation and increased mesangial matrix

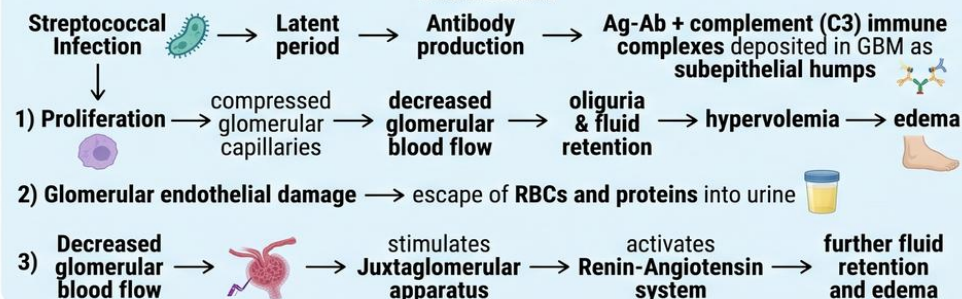


Immunofluorescence: **Lumpy bumpy deposits** of immunoglobulin and complement on the GBM



Electron microscopy: **Electron-dense deposits or humps** on the epithelial side of the GBM

Flowchart



Diagnosis



Urine analysis: **smoky red** or **cola-like** color, oliguria/anuria, high specific gravity, **dysmorphic RBCs**, **RBC casts**, **nephritic range proteinuria**, polymorphonuclear leukocytes. **Serum C3 is reduced** in acute phase, normalizes in 6-8 weeks.

Invasive streptococcal evidence: **positive throat culture** or **rising antibody titers** (ASOT elevated after pharyngeal infection, rarely after skin; **DNase B** best for cutaneous).



Diagnosis likely with: acute nephritic syndrome, recent streptococcal infection evidence, and low C3.

Renal biopsy indications: AKI and RPGN, nephrotic range proteinuria, absence of streptococcal evidence, normal complement levels, or persistent hematuria/proteinuria/diminished renal function/low C3 > 2 months.

Prevention

Early systemic antibiotics do not eliminate GN risk. Culturing and treating native positive family members is recommended.

Treatment

General measures: **bed rest**, **fluid restriction**, **protein/salt restriction**, **high carbohydrate diet**. Specific measures: managing renal insufficiency/hypertension, **10-day course of penicillin** to eradicate streptococci (does not alter GN natural history).

Prognosis

Complete recovery in >95% of children. Recurrences are **extremely rare**.

UNDERSTANDING NEPHROTIC SYNDROME: FROM PROTEINURIA TO TREATMENT

PROTEINURIA: DETECTION & CLASSIFICATION | NEPHROTIC SYNDROME: PATHOGENESIS & DEFINITION | GENERALIZED EDEMA: DIFFERENTIAL DIAGNOSIS & CLINICAL MANIFESTATIONS | TREATMENT & DIAGNOSIS

DETECTION OF PROTEINURIA

Urine Dipsticks Test (Qualitative)



Negative,
Trace (10-20 mg/dL)
1+ (30 mg/dL)
2+ (100 mg/dL)
3+ (300 mg/dL)
4+ (1000-2000 mg/dL)
Detects primarily albuminuria.
Less sensitive for other proteins.

24 Hours Urine Collection (Quantitative)



Normal: ≤ 150 mg/24 hr (≤ 4 mg/m²/hr)
Abnormal: > 4 mg/m²/hr
Nephritic Range: 4-40 mg/m²/hr
Nephrotic Range: > 40 mg/m²/hr

Urine Protein/Creatinine (Upr/UCr) Ratio (Random Specimen)



Normal: < 0.5 (children < 2 yr),
 < 0.2 (children ≥ 2 yr)
Nephritic Range: 0.2-2
Nephrotic Range: > 2

CLASSIFICATION & ETIOLOGY

Transient Proteinuria (Common)



Causes:
Fever, Exercise, Dehydration,
Cold Exposure, Seizure.

Orthostatic (Postural) Proteinuria



Most common persistent proteinuria in school-age. Normal excretion supine, **increased upright (up to 1g/24hr)**. Asymptomatic, no hematuria/hypertension.

Glomerular & Tubulo-Interstitial



Glomerular: Acute Post-Infectious GN, FSGS, Mesangial Proliferative GN, Membranous Nephropathy, MPGN, Lupus Nephritis, IgA Nephropathy, HSP Nephritis, Sickle Cell
Level: < 1 g to > 3 g/24hr

Tubulo-Interstitial: Cystinosis, Wilson, Tubulointerstitial Nephritis, Heavy Metal, ATN, PKD. **Level:** Fixed low-grade (< 1 g/24hr)

Initial Evaluation of Persistent Fixed Proteinuria: Serum Creatinine, 24-hr Urinary Protein, Serum Albumin, C3 Level.

DEFINITION (PRIMARY PEDIATRIC DISORDER)



1. Heavy Proteinuria
(> 40 mg/m²/hr)



2. Hypoalbuminemia
(< 2.5 g/dL)



3. Generalized Edema



4. Hyperlipidemia
(> 250 mg/dL)

ETIOLOGY: Congenital or Acquired (Primary: Idiopathic NS; Secondary: SLE, HSP)

PATHOGENESIS (FLOW DIAGRAM)

Increased GBM Permeability
(Structural/Charge Defect)

PROTEINURIA

HYPOPROTEINEMIA
(Hypoalbuminemia)

HYPERLIPIDEMIA

Low serum proteins stimulate liver protein synthesis (including lipoproteins).
Decreased lipoprotein lipase activity decreases lipid catabolism.
→ Elevates Serum Cholesterol & Triglycerides

GENERALIZED EDEMA

Decreased plasma oncotic pressure → Fluid shifts from intravascular to interstitial tissue.
→ Decreased Intravascular Volume

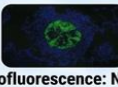
(a) Increased ADH Secretion
→ Water Retention

(b) Decreased Renal Blood Flow → JGA Stimulation
→ Increased Aldosterone Secretion → Sodium & Water Retention (by Kidneys)
→ Further Worsens Generalized Edema

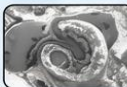
PATHOLOGY (MINIMAL CHANGE DISEASE - MCNS, 85%)



Light Microscopy:
Normal Glomeruli.



Immunofluorescence: Negative.



Electron Microscopy:
Effacement of Epithelial Cell Foot Processes.

GENERALIZED EDEMA: DIFFERENTIAL DIAGNOSIS & CLINICAL MANIFESTATIONS

CLINICAL MANIFESTATIONS



Most common ages 2-6 yr. Presents with **Mild Edema** initially (periorbital, lower extremities). Progresses to **Generalized Anasarca** (ascites, pleural effusions, genital edema). Common symptoms: Anorexia, Irritability, Diarrhea.



DIFFERENTIAL DIAGNOSIS OF GENERALIZED EDEMA



RENAL EDEMA

Starts periorbital (morning). Presents with Ascites. Associated with Hematuria, Oliguria, Hypertension.



CARDIAC EDEMA

Starts in Lower Limbs (sacral if bed-ridden). Presents with Ascites. Signs of Heart Failure (Dyspnea, Orthopnea).



HEPATIC EDEMA

Starts primarily as Ascites. Associated with Liver Cell Failure (Jaundice, Bleeding Tendency).



NUTRITIONAL EDEMA

Starts in Dorsa of Hands & Feet. Associated with Poor Dietetic History, Kwashiorkor features.



ANGIONEUROTIC EDEMA

Localized to Tongue, Lips, Cheeks. Associated with Severe Itching, Recurrent Acute Episodes.

DIAGNOSIS & BIOPSY INDICATIONS



Urine Analysis:
Frothy color, **Heavy Proteinuria** (3+/4+, > 40 mg/m²/hr), Spot P/C Ratio > 2.0 , Waxy/Lipoid/Hyaline Casts, Microscopic Hematuria (20%).



Biochemical Tests:
Serum Albumin < 2.5 g/dL, Total Protein < 4.5 g/dL, **Elevated Cholesterol & Triglycerides**, Normal C3 & C4 Levels.



Indications for Renal Biopsy:
Age < 1 yr or > 10 yr, Hematuria, Hypertension, Renal Insufficiency, Hypocomplementemia, **Steroid Resistance**.

TREATMENT & RESPONSE TO STEROIDS



SUPPORTIVE TREATMENT
Low Sodium Diet, Fluid Restriction (if hyponatremic). IV 25% Human Albumin (0.5g/kg) with Furosemide for Massive Anasarca/Respiratory Distress or Hypovolemic Shock.



CURATIVE TREATMENT
Prednisone at 60 mg/m²/day (max 60 mg) divided for at least 4 consecutive weeks.



RESPONSE CATEGORIES
1) **Steroid Responsive NS:** Complete remission (urine protein trace/negative for 3 consecutive days).
2) **Steroid Dependent:** Relapse on alternate-day therapy OR within 14 days of stopping prednisone.
3) **Frequent Relapsers:** Initial response, but relapses ≥ 4 times in 12 months.
4) **Steroid Resistant:** Complete failure to respond to prednisone within 4-6 weeks of continuous daily use.
Alternative agents: Cyclophosphamide, Cyclosporine, ACE inhibitors.



COMPLICATIONS
Spontaneous Bacterial Peritonitis (S. pneumoniae, E. coli), Thromboembolic Events (Renal vein/sagittal sinus thrombosis due to anti-thrombin III loss/thrombocytosis), Hypovolemic Shock, Acute Renal Failure.

ACUTE KIDNEY INJURY (AKI) - OVERVIEW & MANAGEMENT

1. DEFINITION



RAPID decline (hours to days) in GFR.

Key impairments:

- **IMPAIRMENT** of nitrogenous waste excretion
- **LOSS** of water & electrolyte regulation
- **LOSS** of acid-base regulation



2. GRADING (pRIFLE CRITERIA)



- 1) RISK:** GFR ↓ 25% or UOP < 0.5 ml/kg/h for 8 h
- 2) INJURY:** GFR ↓ 50% or UOP < 0.5 ml/kg/h for 16 h
- 3) FAILURE:** GFR ↓ 75% or UOP < 0.3 ml/kg/h for 24 h or anuric for 12 h
- 4) LOSS:** Persistent failure > 4 weeks
- 5) END STAGE:** ESRD (persistent failure > 3 months)

3. BASIC LABORATORY INVESTIGATIONS



- **URINE VOLUME:** Identify **Oliguria** (UOP < 1 ml/kg/hr or < 400 ml/m²/day) or **Anuria** (< 30 ml/m²/day)
- **RENAL FUNCTION TESTS:** **MARKED INCREASE** in serum Creatinine, BUN, & Urea
- **ACID-BASE EVALUATION:** Document **METABOLIC ACIDOSIS** (low pH, low PaCO₂, low HCO₃)
- **SERUM ELECTROLYTES:** Track **SEVERE HYPERKALEMIA**, dilutional hyponatremia, hypocalcemia, hyperphosphatemia

4. PATHOGENESIS & COMMON CAUSES



PRERENAL (Volume/Perfusion ↓)

- Dehydration
- Hemorrhage
- Sepsis
- Hypoalbuminemia
- Cardiac failure

INTRINSIC RENAL (Structural Damage)

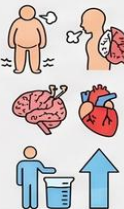
- Glomerulonephritis
- Hemolytic-Uremic syndrome
- Acute tubular necrosis
- Cortical necrosis
- Acute interstitial nephritis

POSTRENAL (Obstruction)

- Posterior urethral valve
- Ureterovesical junction obstruction
- PUJ obstruction
- Urolithiasis
- Neurogenic bladder

CLINICAL MANIFESTATIONS & PHASES

- **MANIFESTATION OF CAUSE.**
- **OLIGURIC PHASE:** Early AKI with Oliguria/Anuria, Edema, Hypertension, Acidotic breathing. **Advanced features:** Uremic (confusion-convulsion-coma), hyponatremic convulsions, hyperkalemic arrhythmias, CHF, GI stress ulcers
- **POLYURIC PHASE:** Indicates early tubular recovery, HIGH VOLUME output



DIFFERENTIAL DIAGNOSIS & TREATMENT

- **PRE-RENAL DIFF:** History volume depletion; SG > 1020, Osm > 500, UNa < 20 mEq/L, FENa < 1%
- **INTRINSIC DIFF:** History nephrotoxins/pharyngitis; SG < 1010, Osm < 350, UNa > 40 mEq/L, FENa > 2%. **Glomerular disease:** RBC/granular casts. **Tubulointerstitial:** WBCs/WBC casts
- **POST-RENAL DIFF:** Palpable bladder mass. **Tests:** Catheterization, Ultrasound, VCUG
- **OTHER TESTS:** CBC, C3, Chest X-ray, Renal biopsy
- **TREATMENT:** Hospitalization, Fluid/Electrolyte management, Treat cause, Correct Hyperkalemia/Acidosis

CHRONIC KIDNEY DISEASE (CKD): DEFINITION, STAGING, ETIOLOGY, INVESTIGATIONS & TREATMENT

Understanding Kidney Structure and Function Abnormalities for Health Implications

1. DEFINITION & STAGING



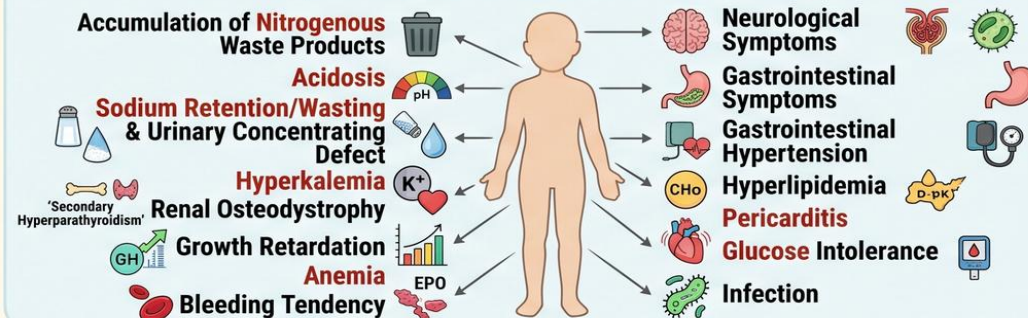
CKD: Abnormalities of kidney structure or function >3 months
OR
Persistent decrease in GFR <60ml/min/1.73m² for >3 months.



STAGING OF CKD



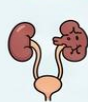
3. CLINICAL & LABORATORY FINDINGS (MANIFESTATIONS & MECHANISMS)



2. DETAILED ETIOLOGY OF CKD IN CHILDREN



<5 YEARS OLD:
Structural/Congenital Anomalies (CAKUT) predominate.



Renal hypoplasia, Renal dysplasia, Reflux nephropathy, Obstructive uropathy (e.g., posterior urethral valves).



Inherited metabolic/genetic disorders (e.g., cystinosis, oxalosis, ARPKD, congenital nephrotic syndrome).



Congenital renal tumors (e.g., Wilms' tumor).



>5 YEARS OLD:
Acquired Glomerular Diseases predominate.



FSGS, MPGN, Chronic glomerulonephritis, HUS sequelae.



Inherited conditions (e.g., Alport's, nephronophthisis, tubulointerstitial nephritis, nephrotoxic drug damage).

4. DIAGNOSTIC INVESTIGATIONS & TREATMENT

INVESTIGATIONS



Blood Chemistry & Monitoring: Hemoglobin, hematocrit, serum electrolytes (K⁺, Na⁺), BUN, serum creatinine (GFR calculation).



Metabolic Bone Mineral Evaluation: Serum Ca, P, Alkaline Phosphatase, Parathyroid Hormone (PTH).



Imaging: Standard long bone X-rays (renal osteodystrophy), Renal ultrasonography (size, structure, obstruction).



Cardiovascular Evaluation: Echocardiography (LVH, cardiac dysfunction).



Urine Analysis: Persistent hematuria, heavy proteinuria, specific gravity. GFR = 0.413 x (length in cm / serum creatinine).

TREATMENT



Fluid/Electrolyte Management: Sodium restriction, loop diuretics.



Growth & Bone Health: Low P diet, phosphate binders, active Vitamin D.



Anemia & Hypertension: rHuEPO. Loop diuretics, ACE inhibitors (enalapril, lisinopril) for proteinuria.



Immunizations & Advanced Care: Standard vaccines (avoid live if immunosuppressed). Drug dosage adjustments.



Renal Replacement Therapy (RRT): Stage 5 CKD (Hemodialysis, Peritoneal Dialysis, Kidney Transplantation).

URINARY TRACT INFECTIONS (UTIs) & PYELONEPHRITIS: A COMPREHENSIVE GUIDE

1. PREVALENCE & ETIOLOGY



UTIs occur in **3-5%** of **GIRLS** and **1%** of **BOYS**.

In **GIRLS**: First UTI often by age 5, peaks during **INFANCY & TOILET TRAINING**.

RECURRENCE: 60-80% within 18 months.

In **BOYS**: Most during **1st YEAR OF LIFE**. Higher risk in **UNCIRCUMCISED** boys.



MALE:FEMALE ratio: 2.8-5.4:1 in 1st year; 1:10 female preponderance **BEYOND 1-2 YEARS**.

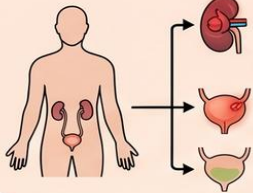


MAIN PATHOGENS: *ESCHERICHIA COLI* (E. coli), followed by *Klebsiella* & *Proteus*. *Staphylococcus saprophyticus* in both sexes. Viral (adenovirus) possible.

IMPORTANT RISK for **RENAL INSUFFICIENCY** or **ESRD**.



2. CLINICAL MANIFESTATIONS & CLASSIFICATION



THREE FORMS: PYELONEPHRITIS (Acute Parenchymal Involvement):

Abdominal/flank pain, **FEVER**, malaise, nausea, vomiting, occasional diarrhea. Newborns/Infants: Nonspecific symptoms (jaundice, poor feeding, irritability, weight loss). Can cause **SCARRING**.

CYSTITIS (Bladder Involvement): Dysuria, **URGENCY**, frequency, suprapubic pain, incontinence, **MALODOROUS** urine. **NO FEVER**, **NO RENAL**.

ASYMPTOMATIC BACTERIURIA: Positive culture, **NO MANIFESTATIONS**. Almost exclusively in **GIRLS**, **BENIGN**, **NO RENAL INJURY**.

3. PATHOGENESIS & RISK FACTORS

PATHOGENESIS: Nearly **ALL** are **ASCENDING INFECTIONS** (Fecal flora to Perineum to Bladder via Urethra). **Rare**: Hematogenous spread.

RISK FACTORS:

- **FEMALE SEX**
- **UNCIRCUMCISED** male
- **VESICoureTERAL REFLEX (VUR)**
- **TOILET TRAINING**
- **VOIDING DYSFUNCTION**



- **urethral instrumentation**
- **TIGHT CLOTHING/UNDERWEAR**
- **PINWORM** infestation
- **CONSTIPATION**
- **LABIAL ADHESION**
- **NEUROPATHIC BLADDER**



4. DIAGNOSIS & IMAGING STUDIES



DIAGNOSIS: URINE CULTURE IS NECESSARY for confirmation.

- Acute renal infection findings: **LEUKOCYTOSIS**, **NEUTROPHILIA**, elevated ESR & CRP.
- **Renal Abscess**: WBC count markedly elevated (20,000-25,000/mm3).
- Blood cultures if sepsis suspected.



IMAGING (Goal: Identify Predisposing Anatomic Abnormalities):

- **RENAL ULTRASONOGRAPHY**: Rules out hydronephrosis, abscess, enlarged kidney (shows acute pyelonephritis in 30-60%).
- **VOIDING CYSTOURETHROGRAM (VCUG)**: Indicated in **ALL children <5 YEARS** with UTI, febrile UTI, school-aged girls with ≥2 UTIs, any male with UTI.
- **MOST COMMON FINDING**: **VESICoureTERAL REFLEX**.
- **DMSA RENAL SCANNING**: Useful if acute pyelonephritis **UNCERTAIN** or to assess **SCARRING**.



5. TREATMENT & PROPHYLAXIS



TREATMENT:

- **ACUTE CYSTITIS**: Prompt treatment to prevent progression. **MILD SYMPTOMS (3-5 days)**: **TRIMETHOPRIM-SULFAMETHOXAZOLE** or **NITROFURANTOIN** (5-7 mg/kg/24 hr, 3-4 divided doses, active against *Klebsiella-Enterobacter*). **AMOXICILLIN** (50 mg/kg/24 hr) is an option.
- **ACUTE FEBRILE INFECTIONS (PYELONEPHRITIS)**: 14-day **BROAD-SPECTRUM** antibiotics. **Parenteral (preferable)**: **CEFTRIAXONE** (50-75 mg/kg/24 hr, max 2g) **OR** **AMPICILLIN** (100 mg/kg/24 hr) w/TH **AMINOGLYCOSIDE** (e.g., **GENTAMICIN** 3-5 mg/kg/24 hr). **MONITOR**: Serum creatinine & trough gentamicin levels daily (OTOTOXICITY & NEPHROTOXICITY risk).
 - **Aminoglycosides** effective anrative against *Pseudomonas*, enhanced by **URINE ALKALINIZATION** (sodium bicarbonate).
 - **Oral 3rd-gen Cephalosporins** (e.g., **CEFIMIXE**) as effective as parenteral ceftriaxone against gram-negatives (except *Pseudomonas*).
- **ABSCESSSES/OBSTRUCTIONS**: Surgical or percutaneous drainage.



PROPHYLAXIS (Against **RE-INFECTION**): **SULFAMETHOXAZOLE-TRIMETHOPRIM** or **NITROFURANTOIN** at **ONE THIRD** therapeutic dose **ONCE DAILY**. **Amoxicillin** or **cephalexin** are also effective.



CHAPTER 04

IV. Emergency Medicine



CARDIOPULMONARY RESUSCITATION (CPR) • SHOCK MANAGEMENT • GLASGOW COMA SCALE • COMA APPROACH • HYPERBILIRUBINEMIA COMPLICATIONS

STEPS OF CARDIOPULMONARY RESUSCITATION (CPR)

DEFINITION: CPR is a sudden unexpected cessation of circulation and or respiration in a patient who is unlikely to die and whom the potentials for recovery are good. In children, cardiac arrest is usually secondary to respiratory failure or shock, unlike adults where it is commonly primary cardiac. Initial rhythm is usually bradycardia progressing to asystole. VF/pulseless VT is less common.



If successful resuscitative efforts are not performed within **5 minutes**, brain or cardiovascular death will occur. Irreversible brain damage happens after about **seven minutes**.

I- Basic Life Support (A, B, C): It aims to restore spontaneous breathing and circulation.

A: Airway control



1) Open the airway by triple airway maneuver

1) **Head tilt:** The head is extended backwards by both hands.
2) **Jaw thrust:** The jaw is pushed forward by the fingers under the angles of the mandible.
3) **Mouth opening:** the mouth is opened by the thumb and index fingers.



2) **Clear the airway**
Remove any obvious foreign bodies cautiously. Avoid finger sweep... Remove the mouth and Oropharynx.



3) **Maintain patent airway**
Oropharyngeal airway is inserted... The appropriate size is the appropriate size.
Endotracheal tube (ETT): In PICU... A laryngoscope is essential... Straight blade preferred up to 8 years (Size 1 newborn/infant, Size 2 child <10yr, Size 3 older child)

B: Breathing support



Mouth-to-mouth breathing when equipment is not available.



Bag and mask ventilation with O₂ in hospital wards. Ventilatory rate is 20 breaths/minute.



Bag and tube ventilation with O₂ in emergency departments or PICU.

C: Circulation support (with cardiac compression)



Site of compression is Midsternum point. **Technique:** In newborns use hand encircling technique; In infants use two fingers technique; In young children use one hand technique; In old children use two hands technique. Depth of compression: 2 cm (infant), 3 cm (young child), 4cm (old child). Frequency of compression is about 100/minute in all ages. Ratio: Continue ventilation and cardiac compression at a ratio of 1:5.

II-Advanced Life Support (D, E, F): Indicated when basic life support was not successful. Maintain A+B+C with: Insert I.V line and give fluids (Ringer's lactate or normal saline: 20 ml/kg over 10 min) which can be repeated.



D: Drugs (I.V.)
Give **sodium bicarbonate:** 1 ml/kg of 8.4 solution or 2 ml/kg of 5% solution. Give **adrenaline:** 0.1 ml/kg of diluted solution (1ml + 9ml saline).



E: ECG monitoring
Important to detect various cardiac arrhythmias. With **asystole:** Repeat adrenaline at 10 times the first dose.



F: Fibrillation control
Indicated with ventricular fibrillation. Defibrillate up to 3 times: 2 joules/kg, 4 joules/kg, and 8 joules/kg.

III-Prolonged life support (G, H, I): Post-resuscitation management aiming for:

G: Recognition and treatment of the causative disease according to the etiology.

H: Brain recovery by control of any convulsive fits or acute increased intracranial pressure.

I: Intensive care for multiple system support done carefully in the PICU.



Resuscitative efforts should continue as long as brain death does not occur. Repeated examination of eye reflexes (pupillary reaction to light) every 5 minutes is important to exclude brain death. Leaders terminate resuscitation based on the presence of brain death.



Confused



Shallow breathing



CO₂ monital status



Slow heart



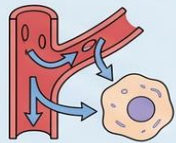
Low BP cuff

Outcomes include death, complete recovery, or partial recovery with post-resuscitation multiple organ system failure due to global ischemia.

Warning signs of impending arrest: altered mental status, decreasing respiratory effort, rising CO₂, bradycardia, and hypotension.

SHOCK: CIRCULATORY INADEQUACY & TISSUE HYPOPERFUSION

Clinical State of Disrupted Oxygen and Nutrient Delivery



Definition: Clinical state of circulatory inadequacy due to disruption of tissue perfusion, leading to inadequate oxygen/nutrient supply and metabolite removal from end-organ cells.

Key Problem: Cellular Hypoxia → **Anaerobic Metabolism** → **Lactic Acidosis** → **Organ Dysfunction** ☠️



In children, **Hypotension** is a **LATE** sign; early recognition is **CRITICAL**.

Oxygen Delivery (DO₂) = Cardiac Output (CO) x Oxygen Content

Cardiac Output = Heart Rate (HR) x Stroke Volume

Children compensate by increasing HR & SVR, but **CANNOT** significantly increase stroke volume, leading to sudden deterioration when compensation fails.

Causes & Types of Shock



1) Septic:
Primary Septicemia (fulminant sepsis), **Secondary Septicemia** (serious focal infection), **Endogenous Septicemia** (bacterial translocation from gut).



2) Hypovolemic:
Severe Dehydration (diarrhea, vomiting, DKA), **Severe Hemorrhage** (traumatic/spontaneous), **Severe Burns** (plasma losses).



3) Obstructive:
Tension Pneumothorax/Hemothorax, **Cardiac Tamponade**, **Critical Obstructive Lesions** (pulmonary/aortic stenosis), **Pulmonary Embolism**.



4) Cardiogenic:
Severe Acute Heart Failure, **Advanced Shock** (myocardial ischemia), **Late Septic Shock** (toxic myocarditis).
Signs: Tachycardia, raised JVP, gallop rhythm, lung crepitations, enlarged tender liver.



5) Kinetic/Distributive:
Anaphylactic Shock (drugs, foods, stings), **Neurogenic Shock** (head/spinal trauma → reduced sympathetic tone), **Early Septic Shock** (blood pooling → hypotension).



6) Metabolic:
Acute Suprarenal Failure (hemorrhage/necrosis).

Clinical Grading:



Grade I (Early Shock):
Peripheral hypoperfusion.
Tachycardia & poor peripheral perfusion.



Grade II (Established Shock):
Arterial hypotension.
Tachycardia, poor perfusion, & hypotension.



Grade III (Advanced Shock):
Vital organ hypoperfusion.
Multiple Organ System Failure (MOSF).



Grade IV (Irreversible Shock):
Irreversible cellular damage.
Refractory metabolic acidosis.

Classification:

Compensated Shock:
Tachycardia, normal/near normal BP, reduced capillary refill, cool peripheries, concentrated urine.



Uncompensated Shock:
Circulatory Compensation Failure, fall in systolic BP, acidosis, oliguria, reduced level of consciousness.

Manifestations of MOSF



Kidneys: Acute Renal Failure (oliguria, acidosis).



Lungs: ARDS (Acute Respiratory Distress Syndrome).



GIT: Ischemia, stress ulcers, hemorrhage, ileus.



Metabolic:
Lactic Acidosis, electrolyte disturbance.



Blood: DIC, thrombocytopenia.



Brain: Hypoxic Ischemic Encephalopathy (disaurea, convulsions).



Heart: Myocardial ischemia, serious arrhythmias.



Liver: Acute Hepatic Failure.

Management



Monitoring:
Clinical tracking (early stages), **Full Laboratory/Invasive tracking** (late stages - ABG, electrolytes, renal functions, CVP).



Cardiovascular Support:
Oxygen Therapy (High-flow O₂, non-rebreather mask, goal SpO₂ ≥ 94%).



Preload Augmentation:
Volume Expander: Isotonic Crystalloid (NS/Ringer's) IV 20 ml/kg over 10-15 mins, repeat up to 60 ml/kg.



Afterload Reduction: IV **Nitroprusside** (0.5-10 ug/kg/min) or **Nitroglycerin** (1-20 ug/kg/min) in cardiogenic shock if no response to inotropes.



Specific Treatment:

- Septic:** Early combined parenteral antibiotics.
- Hypovolemic:** Deficit IV fluids (80-100 ml/kg over 8-10 hrs), whole blood (Deficit = Hb x BW x 6), plasma transfusion for burns.
- Obstructive:** Surgical intervention (thoracocentesis, pericardiocentesis).
- Anaphylactic:** Epinephrine (SC/IV 0.01 mg/kg), Hydrocortisone, Antihistaminic, nebulized Salbutamol.

Complications of Untreated Shock:

Acute Kidney Injury, DIC, ARDS, MODS, Death.



Glasgow Coma Scale (GCS) & its Advantages

Purpose: To assess **level of consciousness and neurological function in children with altered mental status.**
The Glasgow Coma Scale (GCS) provides a very rapid assessment of cerebral cortical function.



👉 **Pediatric GCS** is adapted for verbal and motor responses in children and infants.

These scales should be scored serially over time to detect disease improvement or progression.



Scale Components:

1. Eye Opening (Total points 4) 👁️

- 👁️ • Spontaneous (4)
- 👁️ 🗣️ • To voice (3)
- 👁️ • To pain (2)
- 👁️ • None (1)

2. Verbal Response (Total points 5) 💬

For Older Children:

- 😊 OK • Oriented (5)
- 😕 ? • Confused (4)
- 😬 APPLE... SHOE... • Inappropriate words (3)
- 😬 Moon... grr... • Incomprehensible words (2)
- 😐 • None (1)

For Infants and Young Children:

- 😊 • Appropriate words, smiles, fixes, and follows (5)
- 😭 • Consolable crying (4)
- 😭 • Persistently irritable (3)
- 😭 • Restless, agitated (2)
- 😐 • None (1)

3. Motor Response (Total points 6) 💪

- 👉 STOP • Obeys (6)
- 👉 • Localizes pain (5)
- 👉 • Withdraws (4)
- 👉 • Flexion (3) (Decorticate)
- 👉 • Extension (2) (Decerebrate)
- 👉 • None (1)



Total score ranges from 3 to 15.



Advantages of GCS:



- 1) **Objective & standardized:** Provides a numerical score (3-15) to track consciousness over time.
- 2) **Simple and quick:** can be applied at bedside in emergencies.
- 3) **Widely accepted:** used internationally in trauma, PICU, and emergency settings.
- 4) **Predictive value:** Low GCS (≤ 8) indicates severe brain injury and need for aggressive management, including mechanical ventilation and intracranial pressure monitoring (airway protection).
- 5) **Communication tool:** Allows clinicians to communicate patient status clearly.



UNDERSTANDING COMA: DEFINITION, CAUSES, GRADING, DIAGNOSIS & MANAGEMENT

COMA: DEFINITION

Coma is a state of **prolonged unconsciousness** in which the child **CANNOT BE AROUSED** even with painful stimuli.



CAUSES OF COMA

1) PRIMARY BRAIN LESION

- **Intracranial infection** (Meningitis, encephalitis, brain abscess)
- **Intracranial hemorrhage** (Traumatic head injury or Non-traumatic like hemophilia, DIC, ITP, rupture aneurysm, A-V malformation)
- **Cerebral infarction**
- **Status epilepticus**
- **Brain tumors**

2) SECONDARY BRAIN LESION (ENCEPHALOPATHY)

- **Hypoxic** encephalopathy (associated with hypoxia, shock, severe anemia)
- **Endogenous** encephalopathy (associated with severe dehydration, metabolic acidosis, electrolyte disturbances, acute renal failure, acute hepatic failure, DKA, hypoglycemia, hyperammonemia)
- **Exogenous** encephalopathy (poisoning)

CLINICAL GRADING OF COMA

GRADE I (STUPOR)	The patient can be aroused for only a SHORT PERIOD (less than a minute).	
GRADE II (LIGHT COMA)	The patient CANNOT BE AROUSED by painful stimuli but responds to pain by WITHDRAWAL MOVEMENTS .	
GRADE III (DEEP COMA)	NO RESPONSE to painful stimuli but breathes spontaneously.	
GRADE IV (DEEP COMA WITH APNEA)	NO RESPONSE to painful stimuli and APNEA ; brain death occurs in 5 minutes if not mechanically ventilated.	

LEVEL OF DYSFUNCTION & SIGNS

- **LEVEL OF DYSFUNCTION** (CORTICAL VS BRAIN STEM): Determined by eliciting brain stem reflexes (pupillary reaction, corneal reflex, oculocephalic reflex). In cortical lesions, brain stem reflexes are **INTACT**; in brain stem lesions, reflexes are **SLUGGISH OR LOST**.
- **ASSOCIATED NEUROLOGICAL SIGNS** Convulsions and signs of **ACUTE INCREASED INTRACRANIAL PRESSURE** (bulged anterior fontanel, sluggish pupillary reaction, hypertonia, hyperreflexia).
- **LATERALIZING SIGNS** Asymmetric motor response, asymmetric brain stem reflexes, and **FOCAL CONVULSIONS**.
- **FEATURES OF MENINGEAL IRRITATION** **MENINGEAL IRRITATION: NECK RIGIDITY/RETRACTION**, and positive **KERNIG** and **BRUDZINSKI** signs.

DIAGNOSIS & INVESTIGATIONS

- ✓ History, physical examination, and **URGENT LABORATORY CHECKS**: Arterial blood gases (ABG), serum electrolytes, blood sugar level, renal function (blood urea and creatinine), and **SÉPSIS SCREEN** (CBC, ESR, CRP).
- ✓ Other tests like CSF examination, **CT SCAN**, **MRI**, metabolic screen, or toxic screen depend on clinical suspicion.

MANAGEMENT

- NONSPECIFIC NEUROLOGICAL SUPPORT**
 1. **AIRWAY: OPEN AIRWAY**, suction oral cavity/oropharynx, and insert oropharyngeal airway.
 2. **BREATHING: OXYGEN THERAPY** 40% concentration or mechanical ventilation.
 3. **CIRCULATION: INSERT I.V LINE** and give ringler's lactate or saline, followed by maintenance solutions.
 4. **CONTROL OF CONVULSIONS** and prevention of further fits.
 5. **REDUCTION OF INCREASED INTRACRANIAL PRESSURE: HEAD ELEVATION 30 DEGREES** in neutral position; fluid restriction; diuretics; and mannitol (0.25-0.5 g/kg).
- 6. **CARE OF RESPIRATORY SYSTEM: SUCTIONING**, chest physiotherapy, and nasogastric tube to avoid aspiration.
- 7. **CARE OF GASTROINTESTINAL SYSTEM: ANTACIDS** to prevent stress ulcers and laxatives to prevent constipation.
- 8. **CARE OF EYES AND SKIN: EYE DROPS/OINTMENT**, frequent change of position to prevent bed sores, and utilization of urine bags or catheters.
- 9. **PREVENTION OF INFECTION** using aseptic measures and antibiotics.
- 10. **NUTRITIONAL SUPPORT** via nasogastric tube feeding or total parenteral nutrition.

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- 👁️ 🗣️ • To voice (3)
- 👁️ • To pain (2)
- 👁️ • None (1)

2. Verbal Response (Total points 5) 💬

For Older Children:

- 😊 OK • Oriented (5)
- 😞 ? • Confused (4)
- 😞 APPLE... SHOE... • Inappropriate words (3)
- 😞 Moooo... grrr... • Incomprehensible words (2)
- 😞 • None (1)

For Infants and Young Children:

- 😊 • Appropriate words, smiles, fixes, and follows (5)
- 😞 • Consolable crying (4)
- 😞 • Persistently irritable (3)
- 😞 • Restless, agitated (2)
- 😞 • None (1)

3. Motor Response (Total points 6) 💪

- 👉 STOP • Obeys (6)
- 👉 • Localizes pain (5)
- 👉 • Withdraws (4)
- 👉 • Flexion (3) (Decorticate)
- 👉 • Extension (2) (Decerebrate)
- 👉 • None (1)



Total score ranges from 3 to 15.



Advantages of GCS:



- 1) **Objective & standardized:** Provides a numerical score (3-15) to track consciousness over time.
- 2) **Simple and quick:** can be applied at bedside in emergencies.
- 3) **Widely accepted:** used internationally in trauma, PICU, and emergency settings.
- 4) **Predictive value:** Low GCS (≤ 8) indicates severe brain injury and need for aggressive management, including mechanical ventilation and intracranial pressure monitoring (airway protection).
- 5) **Communication tool:** Allows clinicians to communicate patient status clearly.

